

# Computing the downward-closure of an indexed language

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## Abstract

We prove the first bounds on the size of a finite automaton recognizing the subword-closure of the language of an indexed grammar. Our upper and lower bounds are both triply exponential for non-deterministic automata, and quadruply exponential for deterministic ones. For the upper bounds, we rely on recent advances in semigroup algebra, which let us compute summaries of bounded size for words with respect to a finite semigroup. By replacing stacks with their summaries, we are able to transform an indexed grammar into a context-free one with the same subword-closure, and then apply existing bounds for context-free grammars.

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## 1 Introduction

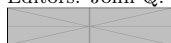
**Downward closures** For finite words  $u$  and  $v$ , we say that  $u$  is a (*scattered*) *subword* of  $v$ , written  $u \preceq v$ , if  $v$  can be obtained from  $u$  by inserting letters. The *downward closure* of a language  $L \subseteq \Sigma^*$  is the set  $L \downarrow = \{u \in \Sigma^* \mid u \preceq v \text{ for some } v \in L\}$  of all subwords of members of  $L$ . The downward-closure of a language—its set of all scattered subwords—is a fundamental abstraction that replaces an arbitrary language by a regular one while preserving information about which patterns may occur in its words. Downward-closures play a central role in verification, model checking, and language-theoretic decision procedures, where they provide effective over-approximations of complex infinite behaviours. Examples of applications include verification of thread-pool systems [3], separability problems with respect to piecewise-testable languages [6], analysis of lossy channel networks [1], and verification of asynchronous shared-memory systems [11].

Higman’s lemma [9] guarantees that the downward-closure of a language is always regular. This fact makes downward-closures a very natural over-approximation method since we are guaranteed that the resulting languages have a simple (although possibly large) representation. However, turning this existential result into concrete constructions on concrete classes of languages with explicit complexity bounds is a long-standing challenge.

**Indexed languages** For context-free grammars, tight upper bounds on the size of automata recognising the downward-closure were proven by Courcelle [5]. However, much less was previously understood for indexed languages, the languages recognised by indexed grammars. Those are a powerful extension of context-free grammars introduced by Aho [2] to capture forms of nested dependencies beyond the reach of pushdown automata. Indexed grammars feature unbounded stacks of stack symbols, making even their structural analysis considerably



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more intricate. Indexed languages can also be defined as the languages recognised by second-order pushdown automata [12], or by Fischer’s macro grammars [7]. In 2015, Zetzsche showed that given an indexed grammar and a finite automaton, one can check whether the language of the latter is the downward-closure of the language of the former [15]. A corollary is that the downward-closure of an indexed language is computable: it suffices to enumerate finite automata until we find one that recognises the downward-closure of the indexed grammar at hand, which happens eventually by Higman’s lemma. However, we cannot infer any bounds from this computability proof, and the size of those automata has been an open question since then.

**Contribution** In this work we establish the first such bounds. Our approach relies on recent advances in finite semigroup theory which provide succinct “summaries” of arbitrarily long words relative to a finite semigroup [8]. By using these summaries to replace indexed stacks, we show how to transform any indexed grammar into a context-free grammar that preserves its downward-closure. Combining this construction with classical bounds for context-free languages yields a triply-exponential upper bound for nondeterministic automata and a quadruply-exponential bound for deterministic automata recognising downward-closures of indexed languages.

We complement these upper bounds with matching lower bounds. For every  $k$ , we construct an indexed grammar of size  $\mathcal{O}(k^2)$  whose downward-closure requires nondeterministic automata of size at least  $2^{2^{2^k}}$  and deterministic ones of size at least  $2^{2^{2^{2^k}}}$ .

Together, these results provide the first complete quantitative picture of downward-closures in the indexed setting, and highlight the surprising power of semigroup-based techniques for analysing higher-order grammar formalisms.

## Related work

TO DO

## Structure of the paper

We recall necessary notations and basic results in Section 2. Then in Section 4 we show how to make a given indexed grammar *sound*, meaning that every partial computation can be extended to a full one. In Section 5 we introduce an object at the core of our construction, the *production semigroup*, and then proceed to define *summaries* of stack contents with respect to this monoid. In Section 8 we leverage those summaries to compute from an indexed grammar a context-free one with the same downward-closure, and infer upper-bounds on the size of finite automata recognizing it. Finally, in Section 9 we complement our upper bounds with matching lower bounds.

*This paper contains internal links, every occurrence of a term is linked to its definition. The reader can click on terms (and some notations), or simply hover over them on some pdf readers, to get their definition.*

## 2 Preliminaries

### 2.1 Words, trees and languages

Given a finite alphabet  $\Sigma$ , we write  $\Sigma^*$  for the set of words over  $\Sigma$ , and  $\Sigma^K$  for the set of words of length  $k$ . Given  $w \in \Sigma^*$ , we denote  $|w|$  its length. We assume familiarity with basic

finite automata theory, see [14] for an introduction.

⌈ We write  $u \preceq v$  if  $u$  is a (scattered) *subword* of  $v$ ; that is, if  $u$  can be obtained from  $v$  by removing some of its letters. The *downward-closure* of a language  $L \subseteq \Sigma^*$  is the set of subwords of words of  $L$ , and is denoted  $L \downarrow$ .

► **Theorem 2.1** (Consequence of [9], Theorem 4.3). *The downward-closure of a language  $L$  is always regular.*

⌈ A finite ordered  $\Lambda$ -labelled tree is a pair  $(\tau, \lambda)$  with  $\tau$  a finite prefix-closed subset of  $\{0, 1\}^*$  such that for all  $\nu \in \tau$ ,  $\nu 1 \in \tau$  implies  $\nu 0 \in \tau$ , and  $\lambda : \tau \rightarrow \Lambda$  a function mapping each node in  $\tau$  to a label. We use the usual terminology for trees, with *node*, *leaf*, *branch*, *subtree*, *parent*, *child*, *ancestor*, *descendant* etc. retaining their usual meaning. The *leaf word* of  $\tau$  is the word obtained by concatenating  $\lambda(\nu_1) \dots \lambda(\nu_k)$ , where  $\nu_1, \dots, \nu_k$  are the leaves of  $\tau$  read from left to right, i.e. in lexicographic order.

## 2.2 Indexed grammars

(Briefly) discuss Chomsky normal form

► **Definition 2.2.** An *indexed grammar* is a tuple of finite sets  $(N, T, I, P, S)$  with

- $N$  the set of non-terminals, and  $S \in N$  the starting non-terminal
- $T$  the set of terminal symbols
- $I$  the set of index symbols, also called stack symbols
- $P$  a set of productions, which are of the following types:

- $A \rightarrow Bf$  with  $A, B \in N$  and  $f \in I$
- $Af \rightarrow B$  with  $A, B \in N$  and  $f \in I$
- $A \rightarrow BC$  with  $A, B, C \in N$
- $A \rightarrow w$  with  $w \in T^*$

We define the *size* of  $\mathcal{G}$  as  $|N| + |P| + \sum_{A \rightarrow w \in P} |w|$ .

A *context-free grammar* (or *CFG*) is an indexed grammar where all productions are of the two last forms.

⌈ A *derivation tree* is a finite ordered tree  $\tau$  whose nodes are labeled by elements of  $NI^* \cup T^*$ , with the following constraints. For each node  $\nu$  with label  $A[z]$  (where  $A \in N$  and  $z \in I^*$ ), exactly one of the following holds

- $\nu$  is a leaf
- $\nu$  has two children labeled  $B[z]$  and  $C[z]$  for some  $A \rightarrow BC \in P$
- $\nu$  has one child labeled  $B[fz]$  for some  $A \rightarrow Bf \in P$
- $\nu$  has one child labeled  $B[z']$  for some  $Af \rightarrow B \in P$ , with  $z = fz'$ .

A derivation tree whose root is labeled  $Az$  and whose leaf word is equal to  $u$  is called a *derivation tree from  $A[z]$  to  $u$* . If  $u \in T^*$  we say that it is a *complete derivation tree*.

⌈ A *sentential form* is a word over the (infinite) alphabet  $NI^* \cup T$ . The set of sentential forms is denoted  $\mathbf{SF}_{\mathcal{G}}$ , or simply  $\mathbf{SF}$  when the grammar is clear from context. An element  $A[z] \in NI^*$  is sometimes referred to as a *term*. The derivation relation  $\Rightarrow_{\mathcal{G}}$  (or simply  $\Rightarrow$ ) over  $\mathbf{SF}$  is defined as:

$$\begin{array}{llll} uA[z]v \Rightarrow uB[z]C[z]v & \text{if } A \rightarrow BC \in P, & uA[z]v \Rightarrow uB[fz]v & \text{if } A \rightarrow Bf \in P, \\ uA[fz]v \Rightarrow uB[z]v & \text{if } Af \rightarrow B \in P, & uA[z]v \Rightarrow uuv & \text{if } A \rightarrow w \in P. \end{array}$$

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for all  $u, v \in \mathbf{SF}$ ,  $A, B, C \in N$ ,  $f \in I$ ,  $z \in I^*$ . We write  $\Rightarrow_{\mathcal{G}}^*$  (or simply  $\Rightarrow^*$ ) for the reflexive transitive closure of  $\Rightarrow_{\mathcal{G}}$ , and the subword relation  $\preceq$  is extended to  $\mathbf{SF}$  in the natural way, i.e.  $u \preceq v$  if  $u$  can be obtained from  $v$  by deleting terms.

Given a sentential form  $u$ , we write  $\mathbf{L}_{\mathbf{SF}}(u)$  for the set  $\{v \in \mathbf{SF} \mid u \Rightarrow^* v\}$ . The language of  $\mathcal{G}$  is denoted  $\mathbf{L}(\mathcal{G})$  and defined as  $\mathbf{L}_{\mathbf{SF}}(S) \cap T^*$ . Furthermore, for all  $X \subseteq N$  and  $u \in \mathbf{SF}$ , the language  $\mathbf{L}_X(u)$  is defined as the set  $\mathbf{L}_{\mathbf{SF}}(u) \cap (X \cup T)^*$ . In particular,  $\mathbf{L}_N(u)$  is the set of sentential forms derivable from  $u$  with all stacks empty, and  $\mathbf{L}_{\emptyset}(u)$  is the set of terminal words which can be derived from  $u$ . If the language  $\mathbf{L}_{\emptyset}(u)$  is non-empty then we say that  $u$  is *productive*.

► **Remark 2.3.** An easy induction shows that for all  $A[z] \in NI^*$ , a sentential form  $u$  can be derived from  $A[z]$  if and only if  $u$  is the leaf word of a derivation tree whose root is labeled  $A[z]$ . In the forthcoming proofs we will either use sentential forms or derivation trees depending on what is most convenient.

► **Remark 2.4.** When describing some examples of indexed grammars, we will use rules of the form  $Af \rightarrow u$  and  $A \rightarrow u$  with  $u \in (N \cup T)^*$ . This is just syntactic sugar, as we can replace them with rules from Definition 2.2 while adding a small number of non-terminals, in the spirit of the Chomsky normal form [4]. This transformation induces a linear blow-up in the size of the grammar. Details are left to Appendix A.

► **Example 2.5.** We can define the language  $L = \{a^n b^{n^2} \mid n \in \mathbb{N}\}$  with an indexed grammar:

$S \rightarrow T g$  #  $g$  is the stack bottom symbol;  
 $T \rightarrow T f$  # we push some number  $n$  of  $f$ ;  
 $T \rightarrow A$   
 $A g \rightarrow \varepsilon$  # if  $n = 0$  then return  $\varepsilon$ ;  
 $A f \rightarrow C$  # if  $n > 0$  we pop an  $f$ ;  
 $C \rightarrow aAB$  # repeat previous and current rule  $n$  times to get  $a^n \prod_{i=0}^{n-1} B f^i g$ ;  
 $B f \rightarrow bbB$   
 $B g \rightarrow b$  # the  $B$ s output  $b^{\sum_{i=0}^{n-1} (2i+1)} = b^{n^2}$ .

► **Example 2.6.** We can define the language  $L = \{w^3 \mid w \in \{a, b\}^*\}$  with an indexed grammar:

$S \rightarrow T g$  #  $g$  is the stack bottom symbol;  
 $T \rightarrow T f_a$   
 $T \rightarrow T f_b$  # we push some word on the stack, as a sequence of  $f_a$  and  $f_b$ ;  
 $T \rightarrow WWW$  # we get three copies of  $W$  with the same stack content;  
 $W g \rightarrow \varepsilon$  # if the stack is empty then return  $\varepsilon$ ;  
 $W f_a \rightarrow aW$   
 $W f_b \rightarrow bW$  # each  $W$  outputs the content of the stack.

► **Remark 2.7.** We can assume without loss of generality that every symbol pushed on the stack carries the information of the production rule used to push it (this property can always be ensured by adding a quantity of new stack symbols and production rules that is at most quadratic in the size of the grammar). Formally, we assume that there are functions  $\alpha, \beta : I \rightarrow N$  such that for every push rule  $A \rightarrow Bf$  we have  $A = \alpha(f)$  and  $B = \beta(f)$ .

### 3 Main results

In this section, we present the main results of this work. Our first main contribution is an algorithm to compute an NFA of at most triply exponential size for the downward closure of an indexed language:

► **Theorem 3.1.** *Given an indexed grammar  $\mathcal{G}$ , one can compute (in triply exponential time) an NFA of at most triply exponential size for  $L(\mathcal{G})\downarrow$ .*

Moreover, we show that the triply exponential upper bound is tight:

► **Theorem 3.2.** *There is a family  $(\mathcal{G}_k)_{k \geq 1}$  of indexed grammars of size linear in  $k$  such that any NFA for  $L(\mathcal{G}_k)\downarrow$  requires at least  $\exp_3(k)$  states.*

**Deterministic automata** Of course, Theorem 3.1 implies a quadruply exponential upper bound for deterministic automata:

► **Corollary 3.3.** *Given an indexed grammar  $\mathcal{G}$ , one can compute (in quadruply exponential time) a DFA of at most quadruply exponential size for  $L(\mathcal{G})\downarrow$ .*

We will also show that the quadruply exponential upper bound is tight:

► **Theorem 3.4.** *There is a family  $(\mathcal{G}_k)_{k \geq 1}$  of indexed grammars of size linear in  $k$  such that any DFA for  $L(\mathcal{G}_k)\downarrow$  requires at least  $\exp_4(k)$  states.*

**Downward closure comparisons** Theorem 3.1 and our technique for Theorem 3.2 allow us to settle the complexity of decision problems related to downward closures of indexed languages. The *downward closure inclusion problem* (for indexed languages) whether two given indexed languages  $L_1, L_2$  satisfy  $L_1\downarrow \subseteq L_2\downarrow$ . Similarly, the *downward closure equivalence problem* asks whether  $L_1\downarrow = L_2\downarrow$ .

In [16, Corollary 18], it was shown that downward closure inclusion and equivalence for indexed languages are co-2-NEXP-hard. Here, we settle their precise complexity:

► **Theorem 3.5.** *Downward closure inclusion and downward closure equivalence for indexed languages are co-3-NEXP-complete.*

### 4 Soundness

Soundness of an indexed grammar expresses the absence of deadlocks: if we can produce a sentential form  $u$ , then from  $u$  we can derive a terminal word in  $T^*$ .

This property is particularly interesting when it comes to downward-closure computation. In a sound indexed grammar, the following property holds: for all  $u \in \mathbf{LSF}(S)$ , if  $v$  is such that  $v \preceq u$ , then  $L_\emptyset(v) \subseteq L_\emptyset(u)\downarrow$ . In other words, we can interleave derivations and deletion of terms without any risk of obtaining “extra” terminal words. This property does not hold in general, because deleting terms that cannot produce terminal words may allow the derivation of a terminal word that is not in the downward closure.

◻ We say that  $\mathcal{G}$  is *sound* if for all  $u \in \mathbf{LSF}(S)$  we have  $L_\emptyset(u) \neq \emptyset$ , that is, from every sentential form obtained from  $S$  we can derive a word in  $T^*$ .

► **Example 4.1.** The following grammar  $\mathcal{G}$  is not sound.

$S \rightarrow ABC$

sound  $\rightarrow$  productive

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203  $S \rightarrow A$   
 204  $A \rightarrow Aa$   
 205  $A \rightarrow \varepsilon$   
 206  $B \rightarrow Bb$   
 207  $B \rightarrow \varepsilon$ .

208 In particular, we have  $B \preceq ABC \in \mathbf{L}_{\mathbf{SF}}(S)$ , but  $b^* = \mathbf{L}_\emptyset(B) \not\subseteq \mathbf{L}_\emptyset(ABC) \downarrow = \emptyset$  (since  $C$  cannot  
 209 produce a terminal word). In this case, it is easy to modify  $\mathcal{G}$  to obtain a sound grammar  
 210 which generates the same language.

211 We will now describe a method to make an indexed grammar sound.

212 ► **Definition 4.2.** For  $X \subseteq N$  and  $z \in I^*$ , we write  $z \cdot X$  for the set  $\{A \in N \mid \mathbf{L}_X(A[z]) \neq \emptyset\}$ .  
 213 Let  $\mathbf{U}$  be defined as the set of non-terminals which can derive a word in  $T^*$ , i.e.

$$214 \quad \mathbf{U} = \{A \in N \mid \mathbf{L}_\emptyset(A) \neq \emptyset\}.$$

215 Note that  $\mathbf{L}(\mathcal{G})$  is empty if and only if  $S \notin \mathbf{U}$ . The following lemma states that the operation  
 216  $z \cdot X$  is a left monoid action of  $I^*$  on the set  $\mathcal{P}(N)$ .

217 ► **Lemma 4.3.** For all  $f \in I$ ,  $z \in I^*$  and  $X \subseteq N$ , we have

$$218 \quad fz \cdot X = f \cdot (z \cdot X).$$

219 **Proof.** If  $A \in f \cdot (z \cdot X)$  then (by definition)  $A[f] \xRightarrow{*} u$ , with  $u \in (z \cdot X \cup T)^*$ . Let  $u^z$  be the  
 220 sentential form obtained by replacing every non-terminal  $B$  in  $u$  with  $B[z]$  (i.e. pushing  $z$   
 221 onto every stack). Since all those non-terminals are in  $z \cdot X$ , there exists  $v \in (X \cup T)^*$  such  
 222 that  $u^z \xRightarrow{*} v$ , implying that  $A[fz] \xRightarrow{*} v$  and so  $A \in fz \cdot X$ .

223 To show the other inclusion, suppose that  $A \in fz \cdot X$  and consider a derivation tree from  
 224  $A[fz]$  to some  $v \in (X \cup T)^*$ . Along every branch there is a first node with a label either in  
 225  $T^*$  or of the form  $B[z]$ . In the latter case we have  $B \in z \cdot X$  (since the tree from this node  
 226 is a derivation tree from  $B[z]$  to an element of  $(X \cup T)^*$ ). After deleting everything below  
 227 these nodes and removing the  $z$  suffix from the stack in each label, we obtain a derivation  
 228 tree from  $A[f]$  to an element of  $(z \cdot X \cup T)^*$ , completing the proof. ◀

229 ► **Lemma 4.4.** For all  $z \in I^*$ ,  $z \cdot \mathbf{U} = z \cdot \emptyset = \{A \in N \mid \mathbf{L}_\emptyset(A[z]) \neq \emptyset\}$ .

**Proof.** Note that the second equality is simply the definition of  $z \cdot \emptyset$ . We proceed by induction  
 on  $z$ . For  $z = \emptyset$ , the statement is equivalent to the definition of  $\mathbf{U}$ . Assuming the statement  
 holds for  $z$ , then two applications of Lemma 4.3 yield

$$fz \cdot \mathbf{U} = f \cdot (z \cdot \mathbf{U}) = f \cdot (z \cdot \emptyset) = fz \cdot \emptyset,$$

230 completing the proof. ◀

231 ◻ The *annotated version* of  $\mathcal{G}$ , denoted  $\overline{\mathcal{G}} = (\overline{N}, T, \overline{I}, \overline{P}, \overline{S})$ , is defined as follows. If  $S \notin \mathbf{U}$   
 232 then it is simply the empty grammar. Otherwise, we have:

- 233 ■  $\overline{N} = \{(A, X) \in N \times 2^N \mid A \in X\}$
- 234 ■  $\overline{I} = I \times 2^N$
- 235 ■  $\overline{P}$  contains the following rules:
- 236 ■  $(A, X) \rightarrow w$  for all  $A \rightarrow w \in P$  with  $w \in T^*$

- 237  $\quad \quad \quad \dashv (A, X) \rightarrow (B, X)(C, X)$  for all  $A \rightarrow BC \in P$  with  $A, B, C \in X$
- 238  $\quad \quad \quad \dashv (A, X) \rightarrow (B, Y)(f, X)$  for all  $A \rightarrow Bf \in P$  with  $Y = f \cdot X$  and  $A \in X$
- 239  $\quad \quad \quad \dashv (A, Y)(f, X) \rightarrow (B, X)$  for all  $Af \rightarrow B \in P$  with  $Y = f \cdot X$ ,  $A \in Y$  and  $B \in X$
- 240  $\quad \quad \quad \dashv \bar{S} = (S, \mathbf{U})$

241  $\lceil$  Finally, we define  $\pi : \bar{N}\bar{I}^* \cup T \rightarrow NI^* \cup T$  to be the morphism projecting each  $(A, X) \in \bar{N}$   
 242 to  $A$ , each  $(f, X) \in \bar{I}$  to  $f$  and each  $a \in T$  to itself.

243 We now define *annotations*. A nonempty stack word  $\bar{z} = (f_n, X_n) \cdots (f_1, X_1) \in \bar{I}^*$  is  
 244 *consistent* if  $X_i = f_i \cdot X_{i-1}$  for all  $i > 1$ ; the empty word is consistent by definition. Note that  
 245 a consistent stack word  $\bar{z}$  is determined by its projection  $\pi(\bar{z})$  onto  $I^*$  and its last element.  
 246 Given  $z = f_n \cdots f_1 \in I^*$  and  $X \subseteq N$ , define  $\bar{z}^X$  as  $(f_n, X_n) \cdots (f_1, X_1)$  with  $X_1 = X$  and  
 247  $X_{i+1} = f_i \cdot X_i$  for all  $i < n$ . Given  $A[z] \in NI^+$  with  $z = f_n \cdots f_1$ , an *X-based annotation*  
 248 of  $A[z]$  is a  $\bar{\mathcal{G}}$  term  $(A, Y)[\bar{z}] = (A, Y)[(f_n, X_n) \cdots (f_1, X_1)] \in \bar{N}\bar{I}^*$  such that  $X_1 = X$ ,  
 249  $Y = f_n \cdot X_n$  and  $\bar{z}$  is consistent. In the case of an empty stack, we say that  $(A, X)$  is an  
 250 *X-based annotation* of  $A$ . In particular, by Lemma 4.3 we have  $X_i = f_{i-1} \cdots f_1 \cdot X$  for all  $i$   
 251 and  $Y = z \cdot X$ . Observe that an *X-based annotation* exists if and only if  $A \in z \cdot X$ , and that  
 252 if it exists it is unique.

253 **► Lemma 4.5.**  $\mathcal{G}$  and  $\bar{\mathcal{G}}$  have the same language, and  $\bar{\mathcal{G}}$  is sound.

254 We prove Lemma 4.5 below, after establishing a preliminary result.

**► Lemma 4.6.** Let  $(A, Y)[\bar{z}] \in \bar{N}\bar{I}^*$  be an *X-based annotation* of  $A[z] \in NI^*$  for some  
 $X \subseteq N$ . If  $u \in L_X(A[z])$ , then there exists  $\bar{u} \in (X \times \{X\} \cup T)^*$  such that

$$\pi(\bar{u}) = u \text{ and } (A, Y)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}.$$

255 In particular, if  $u \in T^*$  then  $A[z] \xrightarrow{*}_{\mathcal{G}} u$  implies that  $(A, X)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} u$  as well.

256 **Proof.** Let us first assume a nonempty stack, setting  $\bar{z} = (f_n, X_n) \cdots (f_1, X_1)$  with  $n \geq 1$   
 257 and  $X = X_1$ . We proceed by induction on the length of the derivation  $A[z] \xrightarrow{*}_{\mathcal{G}} u$ , and  
 258 distinguish cases according to the production rule used in the first step.

- 259  $\dashv$  If the rule is of the form  $A \rightarrow w \in T^*$ , then  $u = w$  and the statement is immediate.
- 260  $\dashv$  If the rule is of the form  $Af \rightarrow B$ , then  $f$  is necessarily equal to  $f_n$ , and we have  
 261  $z = f_n z'$  with  $Bz' \xrightarrow{*}_{\mathcal{G}} u$ . Let  $\bar{z}'$  be such that  $\bar{z} = (f_n, X_n)\bar{z}'$ . Since  $(A, Y)[\bar{z}]$  is an  
 262 *X-based annotation* of  $A[z]$ , we must have  $X_n = z' \cdot X$ , and since  $B[z']$  derives a  
 263 word in  $(X \cup T)^*$  it follows that  $B \in X_n$ . Hence, there is a corresponding derivation  
 264  $(A, Y)[(f_n, X_n)\bar{z}'] \Rightarrow_{\bar{\mathcal{G}}} (B, X_n)[\bar{z}']$ . Moreover,  $(B, X_n)[\bar{z}']$  is a *consistent annotation* of  
 265  $B[z']$ , so the induction hypothesis yields the derivation  $(B, X_n)[\bar{z}'] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}$  for some  $\bar{u} \in$   
 266  $(X \times \{X\} \cup T)^*$  with  $\pi(\bar{u}) = u$ . Hence,  $(A, Y)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}$  as desired. Note that this  
 267 establishes the base case of the induction, since a derivation of length one must consist of  
 268 either this rule or the previous one (and if  $u = B$  then we simply set  $\bar{u} = (B, X)$ ).
- 269  $\dashv$  If the rule is of the form  $A \rightarrow BC$ , then  $u = u_B u_C$  where  $B[z] \xrightarrow{*}_{\mathcal{G}} u_B$  and  $C[z] \xrightarrow{*}_{\mathcal{G}} u_C$ .  
 270 By our assumption, we have  $Y = z \cdot X$ , and since  $A[z]$ ,  $B[z]$  and  $C[z]$  can all produce a  
 271 word in  $(X \cup T)^*$ , we must have  $A, B, C \in Y$  (also,  $A \in Y$  by definition). Consequently,  
 272 we may apply the corresponding derivation in  $\bar{\mathcal{G}}$ :  $(A, Y)[\bar{z}] \Rightarrow_{\bar{\mathcal{G}}} (B, Y)[\bar{z}](C, Y)[\bar{z}]$ . Note  
 273 that  $(B, Y)[\bar{z}]$  and  $(C, Y)[\bar{z}]$  are *X-based annotations* of  $B[z]$  and  $C[z]$  respectively, so  
 274 by the induction hypothesis there are  $\bar{u}_B, \bar{u}_C \in (X \times \{X\} \cup T)^*$  such that  $\pi(\bar{u}_B) = u_B$ ,  
 275  $\pi(\bar{u}_C) = u_C$ ,  $(B, Y)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}_B$  and  $(C, Y)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}_C$ . Setting  $\bar{u} = \bar{u}_B \bar{u}_C$ , we thus have  
 276  $\pi(\bar{u}) = u$  and  $(A, Y)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}$ , as desired.



277 ■ If the rule is of the form  $A \rightarrow Bf$ , then  $B[fz] \xrightarrow{*}_{\mathcal{G}} u$ . As before, we have  $Y = z \cdot X$ ,  
 278 and  $A \in Y$  by definition. Let  $Y' = f \cdot Y = fz \cdot X$ . Since  $B[fz]$  can produce a word in  
 279  $(X \cup T)^*$ , it must be the case that  $B \in Y'$ . Hence, we have the corresponding derivation  
 280  $(A, Y)[\bar{z}] \Rightarrow_{\bar{\mathcal{G}}}(B, Y')[\bar{z}]$ . Note that  $(B, Y')[\bar{z}]$  is an  $X$ -based annotation of  
 281  $B[fz]$ , so from the induction hypothesis we obtain  $\bar{u} \in (X \times \{X\} \cup T)^*$  such that  $\pi(\bar{u}) = u$   
 282 and  $(B, Y')[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}$ . Hence,  $(A, Y)[\bar{z}] \xrightarrow{*}_{\bar{\mathcal{G}}} \bar{u}$ .

283 This concludes the proof for a nonempty stack. In the case where  $z$  is empty, observe that  
 284 only the first, second and fourth rules may be the first step in a derivation, and in those  
 285 cases the arguments all go through unchanged (i.e. with  $Y = X$ ). ◀

286 **Proof of Lemma 4.6.** The inclusion  $L(\bar{\mathcal{G}}) \subseteq L(\mathcal{G})$  is obtained as follows. For all  $w \in L(\bar{\mathcal{G}})$ ,  
 287 we have a derivation tree for  $\bar{\mathcal{G}}$  from  $(S, \mathbf{U})$  to  $w$ . Since  $\pi$  maps the rules in  $\bar{P}$  to rules in  $P$ ,  
 288 it follows that the tree obtained by applying  $\pi$  to each node is a derivation tree from  $S$  to  $w$   
 289 for  $\mathcal{G}$ , implying that  $w \in L(\mathcal{G})$ .

290 To obtain  $L(\mathcal{G}) \subseteq L(\bar{\mathcal{G}})$ , suppose that  $w \in L(\mathcal{G})$ . Then there is a derivation  $S \xrightarrow{*}_{\mathcal{G}} w$ ,  
 291 and since  $(S, \mathbf{U})$  is a  $\mathbf{U}$ -based annotation of  $S$  it follows from Lemma 4.6 that there is a  
 292 derivation  $(S, \mathbf{U}) \xrightarrow{*}_{\bar{\mathcal{G}}} w$ , implying that  $w \in L(\bar{\mathcal{G}})$ .

293  $\square$  It remains to prove soundness. Given  $\bar{u} \in (\bar{N}\bar{T}^* \cup T)^*$ , we say that  $\bar{u}$  is *sound* if every  
 294 term  $(A, X)[\bar{z}]$  of  $\bar{u}$  is a  $\mathbf{U}$ -based annotation of some  $A[z] \in NI^*$ .

295 We wish to show that

- 296 1. every  $\bar{u}$  such that  $(S, \mathbf{U}) \Rightarrow_{\bar{\mathcal{G}}} \bar{u}$  is sound;
- 297 2. for every sound  $\bar{u}$  there exists  $w \in T^*$  such that  $u \xrightarrow{*}_{\bar{\mathcal{G}}} w$ .

298 The first point follows from the definition of  $\bar{\mathcal{G}}$  by an easy induction on the derivation  
 299  $(S, \mathbf{U}) \Rightarrow_{\bar{\mathcal{G}}} \bar{u}$ . For the second point, it is enough to prove it for a single term  $\bar{u} = (A, X)[\bar{z}]$ ,  
 300 since a sentential form can produce a terminal word if and only if all its terms can.

301 As  $(A, X)[\bar{z}]$  is a  $\mathbf{U}$ -based annotation of some  $A[z]$ , then  $X = z \cdot \mathbf{U}$ . Since  $A \in X$  by  
 302 definition of  $\bar{N}$ , there is a derivation from  $A[z]$  to a word of  $T^*$ . As a result, by Lemma 4.6  
 303 there is a derivation from  $(A, X)[\bar{z}]$  to a word in  $T^*$ . ◀

304 ► **Remark 4.7.** Although we are able to turn any indexed grammar into a sound one with  
 305 the same language, this transformation incurs an exponential blow-up in the size. Therefore,  
 306 in order to obtain tight complexity bounds, we do not simply assume soundness of the input  
 307 grammar. Rather, we work with annotated versions explicitly when needed.

## 308 5 The production semigroup

309 Another key aspect of our construction is the **production semigroup**. It is a finite semigroup in  
 310 which we evaluate stack contents. Essentially, we map a stack content  $z \in \bar{T}^*$  to a semigroup  
 311 element which says, for each pair  $A, B$  of non-terminals, whether we can obtain  $B$  from  $A[z]$ .

312 The idea is that since we only want to compute the language of  $\mathcal{G}$  up to downward-closure,  
 313 if we have a derivation from  $A[z]$  to a sentential form  $uBv$ , we can erase  $u$  and  $v$ , thus turning  
 314  $A[z]$  into  $B$ . Even better, if we have a derivation from  $A[z]$  to  $A$  then, roughly speaking,  
 315 this means we can turn any term  $A[zz']$  into  $A[z']$ . We aim to find such patterns in order to  
 316 transmit the flexibility given by the subword-closure to the stack contents.

317 Let us now start by describing formally the semigroup at hand. It is a bit more involved  
 318 than what is described above since we have to account for the soundness issues explained in  
 319 the previous part.



► **Example 5.1.**

TODO

not sure if an example is necessary here?

We first introduce some terminology. A semigroup  $(\mathbf{S}, \cdot)$  is a set equipped with an associative product. We will often identify a semigroup with its set of elements and denote it simply as  $\mathbf{S}$ , when the product operation is clear. A monoid  $(\mathbf{M}, \cdot, \mathbf{1}_{\mathbf{M}})$  is a semigroup with a neutral element  $\mathbf{1}_{\mathbf{M}}$ .

Given a semigroup  $(\mathbf{S}, \cdot)$ , define  $\psi_{\mathbf{S}} : \mathbf{S}^* \rightarrow \mathbf{S}$  to be its *evaluation morphism*, which maps a sequence of elements of  $\mathbf{S}$  to its product. An element  $x \in \mathbf{S}$  is *idempotent* if  $x \cdot x = x$ . The set of idempotent elements of  $\mathbf{S}$  is denoted  $\mathbf{Idem}(\mathbf{S})$ .

In all that follows we fix an indexed grammar  $\mathcal{G} = (N, T, I, P, S)$ . We will mainly work with its annotated version  $\bar{\mathcal{G}}$ . Note that we cannot use the annotation construction as a black box and simply assume that the input grammar is sound (see Remark 4.7).

Define  $\mathbb{M}$  as the boolean matrix monoid over non-terminals of  $\mathcal{G}$ : An element of  $\mathbb{M}$  is a matrix in  $\mathbb{B}^{N \times N}$ , with  $\mathbb{B} = \{\mathbf{true}, \mathbf{false}\}$ . The product is the usual product of matrices over the boolean ring  $(\mathbb{B}, \vee, \wedge)$ , and the neutral element is the matrix with  $\mathbf{true}$  on the diagonal and  $\mathbf{false}$  everywhere else. Given matrices  $M_1, M_2$ , we write  $M_1 M_2$  for their product.

We also define, for all  $X \subseteq N$ , the *reachability relation*  $\mathcal{R}_X$  over  $N$ . It relates two non-terminals if from the first one we can produce a sentential form containing the second one, with empty stacks. Formally,

$$A \mathcal{R}_X B \text{ if and only if there is a derivation } (A, X) \xRightarrow{*}_{\bar{\mathcal{G}}} u(B, X)v \text{ with } u, v \in \mathbf{SF}$$

(note that due to the *soundness* property, it is only important that  $(B, X)$  have an empty stack).

Let  $\alpha, \beta$  be the functions described in Remark 2.7 for  $\mathcal{G}$ .

Define  $\mathbb{S}$  as the following semigroup. An element of  $\mathbb{S}$  is either a tuple  $(B, Y, M, A, X)$  with  $A, B \in N$  non-terminals,  $X, Y \subseteq N$  sets of non-terminals, and  $M \in \mathbb{M}$ , or  $\perp$ . The product operation is defined as follows:

$$(B_2, Y_2, M_2, A_2, X_2) \cdot (B_1, Y_1, M_1, A_1, X_1) = \begin{cases} (B_2, Y_2, M_1 M_2, A_1, X_1) & \text{if } X_2 = Y_1 \text{ and } B_1 \mathcal{R}_{X_2} A_2 \\ \perp & \text{otherwise.} \end{cases}$$

Further, for all  $(B, Y, M, A, X)$  we set  $\perp \cdot (B, Y, M, A, X) = (B, Y, M, A, X) \cdot \perp = \perp \cdot \perp = \perp$ .

We define a morphism  $\varphi : \bar{T}^+ \rightarrow \mathbb{S}$  as follows. For each letter  $(f, X) \in I$ , set

$$\varphi(f, X) = (\beta(f), f \cdot X, M_{f, X}, \alpha(f), X),$$

where for all  $A, B \in N$ ,  $M_{f, X}(A, B) = \top$  if and only if there exist  $u, v \in (X \cup T)^*$  such that  $A[f] \xRightarrow{*}_{\bar{\mathcal{G}}} u B v$ .

We say that a non-empty stack  $\bar{z} = (f_n, X_n) \cdots (f_1, X_1) \in \bar{T}^*$  is *feasible* if there is a derivation  $(\alpha(f_1), X_1) \xRightarrow{*}_{\bar{\mathcal{G}}} u(\beta(f_n), f_n \cdot X_n)[\bar{z}]v$  with  $u, v \in \mathbf{SF}$ .

► **Lemma 5.2.** *Let  $\bar{z} = (f_n, X_n) \cdots (f_1, X_1) \in \bar{T}^*$  be a non-empty stack. The following are equivalent:*

1.  $\bar{z}$  is feasible
2.  $\varphi(\bar{z}) \neq \perp$
3. for all  $i > 1$ ,  $X_i = f_i \cdot X_{i-1}$ ,  $\beta(f_{i-1}) \mathcal{R}_{X_i} \alpha(f_i)$ .

## 23:10 Computing the downward-closure of an indexed language

358 The proof is given in Appendix B

359  $\lceil$  Let  $(A, X) \in \overline{N}$ , we say that  $(A, X)\bar{z}$  is *feasible* if  $\bar{z}$  is feasible,  $X = f_n \cdot X_n$  and  
360  $\beta(f_n) \mathcal{R}_X A$ .

361 The following two lemmas relate  $\mathbb{M}$  and  $\mathbb{S}$  to derivations in  $\mathcal{G}, \overline{\mathcal{G}}$ .

362 **► Lemma 5.3.** *Let  $z \in I^*$ ,  $X \subseteq N$  with  $|z| \geq 1$  and let  $(B, Y, M, A, X) = \varphi(\bar{z}^X)$ , we have,*  
363 *for all  $C \in X$  and  $D \in Y$ ,*

$$\begin{array}{c} 364 \quad C \mathcal{R}_X A \text{ and } B \mathcal{R}_Y D \\ 365 \quad \Updownarrow \\ 366 \quad \text{there exist } u, v \in \mathbf{SF} \text{ such that } (C, X) \xrightarrow{*}_{\overline{\mathcal{G}}} u(D, Y)[\bar{z}^X]v. \end{array}$$

367 This lemma is proven in Appendix C.

368 **► Lemma 5.4.** *Let  $z \in I^*$ ,  $X \subseteq N$  with  $|z| \geq 1$  and let  $(B, Y, M, A, X) = \varphi(\bar{z}^X)$ , we have*

$$\begin{array}{c} 369 \quad M(D, C) = \top \\ 370 \quad \Updownarrow \\ 371 \quad C \in X, D \in Y \text{ and there exist } u, v \in (X \cup T)^* \text{ such that } D[z] \xrightarrow{*}_{\mathcal{G}} uCv. \end{array}$$

372 This lemma is proven in Appendix D.

## 6 Pumping and skipping

374 The *pump rule* lets us push an arbitrary sequence mapping to an idempotent  $e$ . Formally, it  
375 is defined as follows:

$$376 \quad (A, X)[z_e \bar{z}] \rightarrow_{\text{pump}} (A, X)[z'_e z_e \bar{z}]$$

377 for all  $(A, X) \in \overline{N}$ ,  $z_e, z'_e, \bar{z} \in \overline{I}$  and  $e \in \mathbf{Idem}(\mathbb{S})$  such that  $\varphi(z_e) = \varphi(z'_e) = e$ .

378 The *skip rule* is defined as follows:

$$379 \quad (A, X)[z_e v_1 \dots v_N \bar{z}] \rightarrow_{\text{skip}} (A, X)[v_1 \dots v_N \bar{z}]$$

380 for all  $(A, X) \in \overline{N}$ ,  $u_1, \dots, u_N, z_e, \bar{z} \in \overline{I}$  and  $e \in \mathbf{Idem}(\mathbb{S})$  such that  $\varphi(u_1) = \dots = \varphi(u_N) =$   
381  $\varphi(z_e) = e$ .

382 We now show that these additional rule do not increase the downward-closure of the  
383 language of our indexed grammar.

384 **► Lemma 6.1.** *Let  $M \in \mathbb{M}$  a matrix. If  $MM = M$  and we have terms  $A_1, \dots, A_{N+1} \in N$   
385 such that  $M(A_i, A_{i+1}) = \top$  for all  $i$  then there exists  $i$  such that  $M(A_i, A_i) = \top$ .*

386 **Proof.** By pigeonhole principle, there exist  $i < j$  such that  $A_i = A_j$ . Since  $M$  is idempotent,  
387  $M^{j-i} = M$ . Since  $M(A_i, A_{i+1}) = \dots = M(A_{j-1}, A_j) = \top$ , we have  $M(A_i, A_j) = \top$ . Hence  
388  $M(A_i, A_i) = \top$ .  $\blacktriangleleft$

389 **► Lemma 6.2.**  $L^{\text{pump}, \text{skip}}(\overline{\mathcal{G}}) \subseteq L(\overline{\mathcal{G}}) \downarrow$

390 The proof is presented in Appendix G.

## 7 Summarising stack contents

We recall some notions of semigroup algebra which will be necessary in the next section.

Let  $(\mathbf{S}, \cdot)$  be a finite semigroup. Let  $(\mathbf{S}^1, \cdot)$  be the monoid obtained by extending  $\mathbf{S}$  with a neutral element. Formally,  $\mathbf{S}^1 = \mathbf{S} \cup \{1\}$  with  $1 \cdot x = x \cdot 1 = x$  for all  $x \in \mathbf{S}^1$ .

We define the usual Green relations  $\mathcal{J}, \mathcal{L}, \mathcal{R}, \mathcal{H}$  on  $\mathbf{S}$ , starting with the following partial orders:

- $x \leq_{\mathcal{J}} y$  if there exist  $a, b \in \mathbf{S}^1$  such that  $x = a \cdot y \cdot b$
- $x \leq_{\mathcal{L}} y$  if there exist  $a \in \mathbf{S}^1$  such that  $x = a \cdot y$
- $x \leq_{\mathcal{R}} y$  if there exist  $b \in \mathbf{S}^1$  such that  $x = y \cdot b$
- $x \leq_{\mathcal{H}} y$  if  $x \leq_{\mathcal{L}} y$  and  $x \leq_{\mathcal{R}} y$

The relations  $\mathcal{J}, \mathcal{L}, \mathcal{R}, \mathcal{H}$  are the equivalence relations induced by those partial orders: for each  $\mathcal{X} \in \{\mathcal{J}, \mathcal{L}, \mathcal{R}, \mathcal{H}\}$  we define  $\mathcal{X} = \leq_{\mathcal{X}} \cap \geq_{\mathcal{X}}$ , as well as  $<_{\mathcal{X}} = \leq_{\mathcal{X}} \setminus \geq_{\mathcal{X}}$ . We do not include the semigroup  $\mathbf{S}$  in the notation since it will always be clear from the context.

The *regular  $\mathcal{J}$ -length*<sup>1</sup> of  $\mathbf{S}$ , denoted  $JL(\mathbf{S})$ , is defined as

$$JL(\mathbf{S}) = \sup\{k \in \mathbb{N} \mid \exists e_1 >_{\mathcal{J}} \cdots >_{\mathcal{J}} e_k \in \text{Idem}(\mathbf{S})\}.$$

► **Definition 7.1.** The *regular  $\mathcal{J}$ -height* of an element  $x \in \mathbf{S}$  is the maximal number  $h$  such that there exist idempotents  $e_1, \dots, e_h \in E(\mathbf{S})$  such that  $e_1 >_{\mathcal{J}} \cdots >_{\mathcal{J}} e_h \geq_{\mathcal{J}} x$ .

We write  $\text{height}(x)$  for the regular  $\mathcal{J}$ -height of an element  $x \in \mathbf{S}$ . We extend this notation to sequences of elements: Given a sequence of elements  $u \in \mathbf{S}^*$ , we write  $\text{height}(u)$  instead of  $\text{height}(\psi_{\mathbf{S}}(u))$ .

► **Remark 7.2.** Observe that  $\text{height}(x)$  is in  $[1, JL(\mathbf{S})]$  for all  $x \in \mathbf{S}$ . The upper bound follows from the definition of  $JL(\mathbf{S})$ . The fact that  $\text{height}(x)$  is always positive follows from the fact that in a finite semigroup, for all element  $x$  there exists  $n$  such that  $x^n$  is idempotent, hence there is an idempotent  $\leq_{\mathcal{J}}$ -below  $x$ .

In what follows we will extend  $\bar{I}$  with additional letters, one fresh letter  $e^+$  for each idempotent  $e \in \text{Idem}(\mathbf{S})$ . We set  $\varphi(e^+) = e$  for all such  $e$ . Intuitively,  $e^+$  represents a large sequence of factors, which all evaluate to  $e$ .

We extend the regular  $\mathcal{J}$ -height function naturally: Given a sequence of stack symbols  $z \in \bar{I}^*$ , we write  $\text{height}(z)$  instead of  $\text{height}(\varphi(z))$ . Similarly, the regular  $\mathcal{J}$ -height of a sequence of words  $z_1 \dots z_n \in (\bar{I}^*)^*$  is the height of their concatenation, and so on for sequences of sequences of sequences...

► **Definition 7.3.** A *0-summary* is simply the empty word  $\varepsilon$ . Let  $h \in [1, JL(\mathbf{S})]$ .

- An  *$h$ -atom* is a word  $x\sigma$  with  $x \in \mathbb{S}$  and  $\sigma$  an  $h'$ -summary for some  $h' < h$ , so that  $\text{height}(x\sigma) = h$ .
- An  *$h$ -block* is a sequence of the form  $e^+v_1 \cdots v_Nw$  with  $v_1, \dots, v_N$  and  $w$  sequences of  $h$ -atoms and  $e \in \text{Idem}(\mathbf{S})$  such that  $\varphi(v_i) = e$  for all  $i$  and  $\text{height}(e^+v_1 \cdots v_Nw) = h$ .
- An  *$h$ -summary* is a sequence of the form  $\sigma u B_1 \dots B_k$  with  $\sigma$  an  $(h-1)$ -summary,  $u$  a word of  $h$ -atoms and  $B_1, \dots, B_k$   $h$ -blocks, so that  $\text{height}(\sigma u B_1 \dots B_k) = h$ .

<sup>1</sup> Called regular  $\mathcal{D}$ -length in [10]. For finite semigroups,  $\mathcal{D} = \mathcal{J}$ , and we use  $\mathcal{J}$  here since it is more common.

check direction of height/depth.

problem since height is defined by counting idempotents ABOVE

give type of summaries to define height

430 A **summary** is an  $h$ -summary for some  $h$ . The set of summaries is denoted **Summaries**.

431  $\lceil$  We define the following operation on summaries: Given  $h \in [1, JL(\mathbb{S})]$ , an element  $x \in \mathbb{S}$   
 432 and an  $h$ -summary  $\sigma = \sigma' u B_1 \dots B_k$ , we define **push**( $x \blacktriangleright \sigma$ ) as follows:

433 1. If **height**( $x\sigma$ )  $> h$  then let  $h_+ = \mathbf{height}(x\sigma)$ ; **push**( $x \blacktriangleright \sigma$ ) is the  $h_+$ -summary made of  
 434 a single  $h_+$ -atom  $x\sigma$

435 2. Otherwise we have **height**( $x\sigma$ )  $\leq h$ . Since **height**( $x\sigma$ )  $\geq \mathbf{height}(\sigma) = h$ , we even have  
 436 **height**( $x\sigma$ )  $= h$ .

437 a. If **height**( $x\sigma'$ )  $< h$  then **push**( $x \blacktriangleright \sigma$ ) = (**push**( $x \blacktriangleright \sigma'$ )) $u B_1 \dots B_k$

438 b. Otherwise, we have **height**( $x\sigma'$ )  $= h$  (since **height**( $x\sigma'$ )  $\leq \mathbf{height}(x\sigma) \leq h$ ), hence  
 439  $x\sigma'$  is an  $h$ -atom.

440 i. If the sequence of  $h$ -atoms  $(x\sigma')u$  is of the form  $v_0 \dots v_N w$  with  $\varphi(v_i) = e$  for all  
 441  $i$ , for some  $e \in \mathbf{Idem}(\mathbb{S})$ , then define the  $h$ -block  $B = e^+ v_1 \dots v_N w$  (note that  $e^+$   
 442 replaces the first factor  $v_0$ ).

443 A. If there exists  $j$  such that  $B_j$  is of the form  $e^+ v'_1 \dots v'_N w'$  and moreover we have  
 444  $\varphi(v_1 \dots v_N B_1 \dots B_{j-1} u'_1 \dots u'_N e) = e$  then we set **push**( $x \blacktriangleright \sigma$ ) =  $\sigma' B_j B_{j+1} \dots B_k$ .  
 445 If there are several such  $j$  we pick the maximal one.

446 B. Otherwise **push**( $x \blacktriangleright \sigma$ ) =  $B B_1 \dots B_k$

447 ii. Otherwise **push**( $x \blacktriangleright \sigma$ ) =  $u' B_1 \dots B_k$  with  $u' = (x\sigma')u$

448  $\blacktriangleright$  **Example 7.4.**

449 **TODO**

450 We also define the inverse relation: for all  $x \in \mathbb{S}$ ,  $\sigma, \sigma'$  summaries,  $\sigma \in \mathbf{pop}(x \blacktriangleleft \sigma')$  if  
 451 and only if  $\sigma' = \mathbf{push}(x \blacktriangleright \sigma)$ . Given  $\bar{z} \in \bar{I}^*$  we write **push**( $\bar{z} \blacktriangleright \sigma$ ) and **pop**( $\bar{z} \blacktriangleleft \sigma$ ) instead of  
 452 **push**( $\varphi(\bar{z}) \blacktriangleright \sigma$ ) and **pop**( $\varphi(\bar{z}) \blacktriangleleft \sigma$ ).

453  $\lceil$  The **size** of an  $h$ -summary (resp.  $h$ -atom,  $h$ -block) is defined recursively naturally as the  
 454 sum of the sizes of its elements, the size of a single letter being 1. Although summaries can  
 455 have a priori unbounded sizes, we can show that the sizes of the ones we can actually obtain  
 456 by pushing a sequence of letters is exponentially bounded in  $N$ . To prove this, we rely on  
 457 two results of Jecker [10], recalled in Appendix E:

- 458  $\blacksquare$  One guarantees that in a word of elements of exponential length in  $JL(\mathbb{S})$  we can find large  
 459 sequences of consecutive factors that all map to the same idempotent (Theorem E.3).
- 460  $\blacksquare$  Another one bounds the regular  $\mathcal{J}$ -length of a boolean matrix monoid by a polynomial  
 461 in its dimension (see Theorem E.4).

462 We can prove that, when considering  $(h - 1)$ -summaries as single letters, we can bound  
 463 the size of a  $h$ -summary by an exponential in  $N$ . The bound on summaries results from the  
 464 product of those (polynomially many) exponential functions. This proof is heavily inspired by  
 465 by the proof of Theorem 26 in [8].

466  $\blacktriangleright$  **Lemma 7.5.** For all  $z \in \bar{I}^*$ , **push**( $z \blacktriangleright \varepsilon$ ) is of size at most exponential in  $N$ .

467 This lemma is proven in Appendix E.

## 8 A CFG for the downward-closure of the indexed grammar

### 8.1 Definition of the grammar and first inclusion

We define a context-free grammar  $\mathcal{C}_G$  which over-approximates  $\overline{\mathcal{G}}$  by replacing the stack contents with their summaries in  $\mathbb{S}$ . We will see that while the language of  $\mathcal{C}_G$  is in general larger than the one of  $\overline{\mathcal{G}}$ , their downward-closures are the same.

To begin with, let us restrict the set of summaries to the ones that can actually result from a derivation of the indexed grammar.

⌈ A triple  $(A, X, \sigma)$  is *feasible* if  $\text{push}(\bar{z} \blacktriangleright \varepsilon) = \sigma$  for some feasible  $\bar{z}$ . The set of feasible triples is denoted **FT**.

Define the following context-free grammar  $\mathcal{C}_G$ : Its set of non-terminals is **FT**, with  $(S, \mathbf{U}, \varepsilon)$  the initial one. The set of terminal symbols is  $T$ , and the production rules are as follows:

- If  $(A, X) \rightarrow (B, Y)(C, Z) \in \overline{P}$  then  $(A, X, \sigma) \rightarrow (B, Y, \sigma)(C, Z, \sigma)$ , for all  $\sigma \in \mathbf{FT}$ .
- If  $(A, X) \rightarrow (B, Y)(f, X) \in \overline{P}$  then  $(A, X, \sigma) \rightarrow (B, Y, \text{push}((f, X) \blacktriangleright \sigma))$ , for all  $\sigma$  such that  $(A, X, \sigma) \in \mathbf{FT}$  (which implies  $(B, Y, \text{push}((f, X) \blacktriangleright \sigma)) \in \mathbf{FT}$ , as is easily verified).
- If  $(A, X)(f, X) \rightarrow (B, Y) \in \overline{P}$  then  $(A, X, \sigma) \rightarrow (B, Y, \sigma')$ , for all  $\sigma \in \mathbf{FT}$  and  $\sigma' \in \text{pop}((f, X) \blacktriangleleft \sigma)$  such that  $(B, Y, \sigma') \in \mathbf{FT}$ .

Our main theorem is that the downward-closures of the languages of the indexed grammar and this context-free grammar are the same.

► **Theorem 8.1.** *The language of  $\mathcal{C}_G$  has the same downward-closure as  $L(\mathcal{G})$ .*

One of the directions is quite easy: we can simply show that the language of  $\mathcal{C}_G$  contains the one of  $\mathcal{G}$ . This is natural as  $\mathcal{C}_G$  is built as an over-approximation of  $\mathcal{G}$ .

► **Proposition 8.2.**  $L(\overline{\mathcal{G}}) \subseteq L(\mathcal{C}_G)$ .

**Proof.** Let  $(A, X)[z]$  be a feasible term and  $w \in T^*$  and  $(A, X)[z] \xRightarrow{*}_{\overline{\mathcal{G}}} w$ . We show that  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w$ , by induction on the derivation. Note that since  $(A, X)z$  is feasible,  $(A, X, \text{push}(z \blacktriangleright \varepsilon))$  is indeed a non-terminal of  $\mathcal{C}_G$ . We separate cases according to the first rule applied in the derivation  $(A, X)[z] \xRightarrow{*}_{\overline{\mathcal{G}}} w$ .

- If it is of the form  $(A, X) \rightarrow w \in T^*$  then we have  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \rightarrow w$  in  $\mathcal{C}_G$ , hence  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w$ .
- If it is of the form  $(A, X) \rightarrow (B, X)(C, X)$  then we have  $w_B, w_C \in T^*$  such that  $w = w_B w_C$  and  $(B, X)[z] \xRightarrow{*}_{\mathcal{G}} w_B$  and  $(C, X)[z] \xRightarrow{*}_{\mathcal{G}} w_C$ .  
By induction hypothesis,  $(B, X, \text{push}(z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w_B$  and  $(C, X, \text{push}(z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w_C$ .  
Further, we have  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \rightarrow (B, X, \text{push}(z \blacktriangleright \varepsilon))(C, X, \text{push}(z \blacktriangleright \varepsilon))$  in  $\mathcal{C}_G$ , hence  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \Rightarrow_{\mathcal{C}_G} (B, X, \text{push}(z \blacktriangleright \varepsilon))(C, X, \text{push}(z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w_B w_C = w$ .
- If it is of the form  $(A, X) \rightarrow (B, Y)(f, X)$  then we have  $(B, Y)[(f, X)z] \xRightarrow{*}_{\mathcal{G}} w$ . Since  $(A, X)[z]$  is feasible, so is  $(B, Y)[(f, X)z]$ .  
By induction hypothesis,  $(B, Y, \text{push}((f, X)z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w$ . Furthermore, we have  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \rightarrow (B, Y, \text{push}((f, X) \blacktriangleright \text{push}(z \blacktriangleright \varepsilon)))$  in  $\mathcal{C}_G$ , and  $\text{push}((f, X) \blacktriangleright \text{push}(z \blacktriangleright \varepsilon)) = \text{push}((f, X)z \blacktriangleright \varepsilon)$  by definition.  
Therefore,  $(A, X, \text{push}(z \blacktriangleright \varepsilon)) \Rightarrow_{\mathcal{C}_G} (B, Y, \text{push}((f, X)z \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w$ .
- If it is of the form  $(A, Y)(f, X) \rightarrow (B, X)$  then we have  $(B, X)z' \xRightarrow{*}_{\mathcal{G}} w$  with  $z'$  such that  $z = (f, X)z'$ . Since  $(A, Y)[(f, X)z']$  is feasible, so is  $(B, X)[z']$ . By induction hypothesis,  $(B, X, \text{push}(z' \blacktriangleright \varepsilon)) \Rightarrow_{\mathcal{C}_G} w$ .

Furthermore, we have  $\text{push}(z \blacktriangleright \varepsilon) = \text{push}((f, X) \blacktriangleright \text{push}(z' \blacktriangleright \varepsilon))$  by definition and therefore  $\text{push}(z' \blacktriangleright \varepsilon) \in \text{pop}((f, X) \blacktriangleleft \text{push}(z \blacktriangleright \varepsilon))$ .  
 As a consequence,  $(A, Y, \text{push}(z \blacktriangleright \varepsilon)) \rightarrow (B, X, \text{push}(z' \blacktriangleright \varepsilon))$  in  $\mathcal{C}_G$ , and therefore  $(A, Y, \text{push}(z \blacktriangleright \varepsilon)) \Rightarrow_{\mathcal{C}_G} (B, X, \text{push}(z' \blacktriangleright \varepsilon)) \xRightarrow{*}_{\mathcal{C}_G} w$ .

It suffices to apply this property to  $(S, \mathbf{U})$ : for all  $w \in T^*$ , if  $w \in L(\mathcal{G})$  then  $(S, \mathbf{U}) \xRightarrow{*}_{\mathcal{G}} w$ , which implies  $(S, \mathbf{U}, \varepsilon) \xRightarrow{*}_{\mathcal{C}_G} w$ , and thus  $w \in L(\mathcal{C}_G)$ .  $\blacktriangleleft$

## 8.2 Simulating $\mathcal{C}_G$ with pumps and skips

For the other direction, we will simulate an execution of  $\mathcal{C}_G$  by replacing summaries with actual stacks, where symbols  $e^+$  are replaced with a sequence of many factors of stack symbols evaluating to  $e$ . We will use the pump rule and skip rule defined above to extend and reduce sequences of factors mapping to the same idempotent. This will allow us to simulate derivations of  $\mathcal{C}_G$  with  $\overline{\mathcal{G}}$  by adjusting the factors corresponding to symbols  $e^+$ .

We will now show that every word resulting from a derivation of  $\mathcal{C}_G$  is a subword of a word obtained by a derivation with pump and skip in  $\overline{\mathcal{G}}$ . We simply mimic a derivation of  $\mathcal{C}_G$  by applying the same rules, with some pump rules and skip rules to adjust the sequences of idempotents corresponding to letters  $e^+$  in  $\mathcal{C}_G$ .

**Definition 8.3.** Let  $K \in \mathbb{N}$ ,  $z \in \overline{T}^*$ , we say that  $z$  is an **expansion** of an  $h$ -atom /  $h$ -block /  $h$ -summary if the following conditions are satisfied. Those conditions are defined inductively on  $h$ . For  $h = 0$ , we say that  $z$  is an **expansion** of a 0-summary (i.e.,  $\varepsilon$ ) if  $z = \varepsilon$ .

- $z$  is an **expansion** of an  $h$ -atom  $a\sigma'$  if  $z = az'$  with  $z'$  an **expansion** of  $\sigma'$ .
- $z$  is an **expansion** of  $e^+$  if  $z = z_1 \cdots z_p$  with  $\varphi(z_i) = e$  for all  $i$  and for all realisation  $z'$  of  $e$  such that  $|z'| \leq K$ , there exists  $i$  such that  $z_i = z'$ .
- $z$  is an **expansion** of a sequence of  $h$ -atoms  $\alpha_1 \cdots \alpha_m$  if  $z = z_1 \cdots z_m$  with  $z_i$  an **expansion** of  $\alpha_i$  for all  $i$ .
- $z$  is an **expansion** of an  $h$ -block  $e^+v_1 \cdots v_Nw$  if  $z = z_e z^v z^w$  with  $z^v$  (resp.  $z^w$ ,  $z_e$ ) an **expansion** of  $v_1 \cdots v_N$  (resp.  $w$ ,  $e^+$ ).
- $z$  is an **expansion** of an  $h$ -summary  $\sigma'uB_1 \cdots B_m$  if  $z = z^{\sigma'} z^u z_1 \cdots z_m$  with  $z^{\sigma'}$  an **expansion** of  $\sigma'$ ,  $z^u$  an **expansion** of  $u$  and  $z_i$  an **expansion** of  $B_i$  for all  $i$ .

explain

**Lemma 8.4.** Let  $(A, X, \sigma_1) \rightarrow (B, Y, \sigma_2)$  be a rule of  $\mathcal{C}_G$  with  $\sigma_2 \in \text{pop}((f, Y) \blacktriangleleft \sigma_1)$ . Let  $\bar{z}_1$  an **expansion** of  $\sigma_1$ . Then there exists  $\bar{z}_2$  an **expansion** of  $\sigma_2$  such that  $(A, X)[\bar{z}_1](\rightarrow_{\text{pump}} + \rightarrow_{\text{skip}})^*(A, X)[(f, Y)\bar{z}_2]$ .

This lemma is proven in Appendix F.

**Proposition 8.5.**  $L(\mathcal{C}_G) \subseteq L^{\text{pump}, \text{skip}}(\overline{\mathcal{G}}) \downarrow$ .

**Proof.** We prove the following statement:

For all  $K \in \mathbb{N}$ , for all non-terminal  $(A, X, \sigma)$  of  $\mathcal{C}_G$  and  $w \in T^*$ , if there exists a derivation of size  $p$  from  $(A, X, \sigma)$  to  $w$  then for all  $D^p$ -expansion  $z$  of  $\sigma$ , there is a derivation *with pump and skip* from  $(A, X)z$  to  $w'$  in  $\overline{\mathcal{G}}$  with  $w \preceq w'$ .

We show this by induction on  $K$ , and distinguish cases according to the rule used in the first step of the derivation of  $w$  in  $\mathcal{C}_G$ .

- 551 ■ If the rule is of the form  $(A, X, \sigma) \rightarrow w$  then we can apply the corresponding rule  
 552  $(A, X) \rightarrow w$  from  $(A, X)z$ , hence  $(A, X)z \Rightarrow_{\bar{\mathcal{G}}} w$ .
- 553 ■ If the rule is of the form  $(A, X, \sigma) \rightarrow (B, X, \sigma)(C, X, \sigma)$  then there exist  $w_b, w_c$  such that  
 554  $w = w_B w_c$  and  $(B, X, \sigma) \xRightarrow{*}_{\mathcal{C}_{\mathcal{G}}} w_B$  and  $(C, X, \sigma) \xRightarrow{*}_{\mathcal{C}_{\mathcal{G}}} w_C$ . We can apply the corresponding  
 555 rule  $(A, X) \rightarrow (B, X)(C, X)$  from  $(A, X)z$  to obtain  $(B, X)z(C, X)z$ . By induction  
 556 hypothesis, since  $z$  is a  $D^p$ -expansion of  $\sigma$ , we have  $(B, X)z \xRightarrow{*}_{\mathcal{C}_{\mathcal{G}}} w'_B$  and  $(C, X)z \xRightarrow{*}_{\mathcal{C}_{\mathcal{G}}} w'_C$   
 557 with  $w_B \preceq w'_B$  and  $w_C \preceq w'_C$ . As a result,  $(A, X, \sigma) \xRightarrow{*}_{\mathcal{C}_{\mathcal{G}}} w'_B w'_C$ , which yields the result  
 558 since  $w = w_B w_C \preceq w'_B w'_C$ .
- 559 ■ If the rule is of the form  $(A, X, \sigma) \rightarrow (B, Y, \text{push}((f, X) \blacktriangleright \sigma))$ , then by Lemma ?? there  
 560 exists  $z'$  a  $D^p$ -expansions of  $\text{push}((f, X) \blacktriangleright \sigma)$  such that  $(B, Y)[z']$  is reachable and  
 561  $L_{\emptyset}((B, Y)[z']) \subseteq L_{\emptyset}((B, Y)[(f, X)z]) \downarrow$ . By Remark ??,  $z'$  is also a  $D^{p-1}$ -expansion of  
 562  $\text{push}((f, X) \blacktriangleright \sigma)$ . By induction hypothesis, there exists  $w' \in L_{\emptyset}((B, Y)[z'])$  such that  
 563  $w \preceq w'$ . Since  $L_{\emptyset}((B, Y)[z']) \subseteq L_{\emptyset}((B, Y)[(f, X)z])$ , we have  $w' \in L_{\emptyset}((A, X)z)$  and thus  
 564 the induction step is proven.
- 565 ■ If the rule is of the form  $(A, X, \sigma) \rightarrow (B, Y, \sigma')$  with  $\sigma = \text{push}((f, Y) \blacktriangleright \sigma')$ , then by  
 566 Lemma ?? there exists  $z'$  a  $D^{p-1}$ -expansions of  $\sigma'$  such that  $(B, Y)[z']$  is reachable  
 567 and  $L_{\emptyset}((A, X)[(f, Y)z']) \subseteq L_{\emptyset}((A, X)[z])$ . By induction hypothesis, there exists  $w' \in$   
 568  $L_{\emptyset}((B, Y)[z'])$  such that  $w \preceq w'$ . As a consequence,  $w' \in L_{\emptyset}((A, X)[(f, Y)z'])$ , which is  
 569 included in  $L_{\emptyset}((A, X)[z])$ . The induction step is proven.

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570 We have proven the induction. To obtain the lemma, let  $w \in L(\mathcal{C}_{\mathcal{G}})$ . There is a derivation in  
 571  $\mathcal{C}_{\mathcal{G}}$  from  $(S, \mathbf{U}, \varepsilon)$  to  $w$ , let  $p$  be its size. Since  $\varepsilon$  is a  $D^p$ -expansion of  $\varepsilon$  and  $(S, \mathbf{U})$  is reachable  
 572 in  $\bar{\mathcal{G}}$ , there is a derivation from  $(S, \mathbf{U})$  to some  $w'$  with  $w \preceq w'$ . As a result,  $w \in L(\bar{\mathcal{G}}) \downarrow$ . ◀

### 573 8.3 Main result

574 We can conclude our proof of the other inclusion.

575 ► **Proposition 8.6.**  $L(\mathcal{C}_{\mathcal{G}}) \subseteq L(\bar{\mathcal{G}}) \downarrow$ .

576 **Proof.** This is a direct corollary of Propositions ?? and ??. ◀

577 By Theorem ??, the size of a summary is bounded by an exponential in the size of  $\mathcal{G}$ . As  
 578 a consequence, the size of  $\mathcal{C}_{\mathcal{G}}$  is at most doubly exponential in the size of  $\mathcal{G}$ .

579 ► **Theorem 8.7.** *For every indexed grammar  $\mathcal{G}$  there exists a context-free grammar  $\mathcal{C}_{\mathcal{G}}$  such*  
 580 *that  $L(\mathcal{C}_{\mathcal{G}}) \downarrow = L(\mathcal{G}) \downarrow$  and  $\mathcal{C}_{\mathcal{G}}$  has double-exponential size in the size of  $\mathcal{G}$ .*

581 We can then compose this bound with the known bound on downward-closure of context-  
 582 free languages to obtain our result.

583 ► **Theorem 8.8** ([5]). *For all CFG  $\mathcal{C}$  there is an NFA  $\mathcal{A}_{\mathcal{C}}$  such that  $L(\mathcal{A}_{\mathcal{C}}) = L(\mathcal{C}) \downarrow$  and  $\mathcal{A}_{\mathcal{C}}$*   
 584 *has exponentially many states in the size of  $\mathcal{C}$ .*

585 Combining the two theorems above, we obtain our upper bound.

586 ► **Theorem 8.9.** *For every indexed grammar  $\mathcal{G}$  there exists a non-deterministic finite*  
 587 *automaton  $\mathcal{A}_{\mathcal{G}}$  such that  $L(\mathcal{A}_{\mathcal{G}}) = L(\mathcal{G}) \downarrow$  and  $\mathcal{A}_{\mathcal{G}}$  has triple-exponentially many states in the*  
 588 *size of  $\mathcal{G}$ .*



## 9 Lower bound

In this section, we prove the lower bounds in our main results. The key ingredient is the following:

► **Proposition 9.1.** *Given  $k$ , we can compute in polynomial time an indexed grammar—of size polynomial in  $k$ —for the language  $\{a^{\exp_3(k)}\}$ .*

This implies Theorem 3.2: An NFA for  $\{a^{\exp_3(k)}\}^\downarrow$  clearly requires at least  $\exp_3(k)$  states. Let  $k \in \mathbb{N}$ .

**Overview** The overall strategy behind Proposition 9.1 is to construct a grammar whose (unique) derivation tree contains a full binary tree of doubly exponential depth: clearly, such a tree must have triply exponentially many leaves. Here, the challenge is to ensure that the paths have doubly exponential length.

Note that by using the stack in each non-terminal, a standard construction allows us to enforce *singly* exponentially long paths: One uses a stack with a binary alphabet of height  $k$ , and then counts up from  $0^k$  to  $1^k$ , resulting in exponentially many steps. This works because when incrementing a binary expansion of the form  $1^\ell 0w$  (the least significant digit being on the left), we need to replace the prefix  $1^\ell 0$  with  $0^\ell 1$ . Here, one stores the number  $\ell \leq k$  in order to restore the stack height  $k$  when pushing  $0^\ell 1$ .

Doing the same for stack height  $2^k$  is not so easy: To restore a stack height of  $2^k$ , we would need to remember (in the non-terminal) a number  $\ell \leq 2^k$ . In fact, enforcing a single run of length  $2^{2^k}$  in a pushdown automaton of polynomial size is not possible.

Instead, we exploit the fact that an indexed grammar can simulate a pushdown automaton *with alternation*: We implement binary counting on a stack of height  $2^k$ ; to make sure that when replacing a prefix  $1^\ell 0$  with  $0^\ell 1$ , we essentially non-deterministically push 0's, but then use alternation to ensure that the produced stack has height exactly  $2^k$ .

**Step I: Checking the stack via alternation** To this end, we introduce syntactic sugar in indexed grammars. We will use rules of the form

$$A \xrightarrow{\mathcal{A}_1, \dots, \mathcal{A}_r} B, \tag{1}$$

where  $\mathcal{A}_1, \dots, \mathcal{A}_m$  is a list of DFAs over the stack alphabet  $I$ . Such a rule has the same effect as a rule  $A \rightarrow B$ , but it can only be applied to an occurrence  $A[z]$  if for each  $i = 1, \dots, m$ , the word  $z$  has a prefix in  $L(\mathcal{A}_i)$ . Such rules can be implemented with only polynomial overhead: Introduce non-terminals  $C_i, D_i$  for each  $i = 1, \dots, r$  and also  $E_q$  for each state  $q$  in the DFAs  $\mathcal{A}_1, \dots, \mathcal{A}_r$  (we assume the state sets are disjoint). Then we can simulate (1) using  $A \rightarrow D_1 C_1$ ,  $D_i \rightarrow D_{i+1} C_{i+1}$  for  $i = 1, \dots, r-1$ , and  $D_r \rightarrow B$ , which split the node  $A[z]$  into nodes  $B[z]$  and  $C_1[z], \dots, C_r[z]$ . We then run  $\mathcal{A}_i$  using rules  $C_i \rightarrow E_q$ , where  $q_0$  is the initial state of  $\mathcal{A}_i$ , for each  $i$ . The non-terminals  $E_q$  simulate the DFAs: for each transition  $(p, a, q)$ , we have  $E_p a \rightarrow E_q$ . To check acceptance, we have  $E_q \rightarrow \varepsilon$  for each final state  $q$ .

**Step II: Implementing a binary counter in DFAs** We want to use rules (1) to check that the current stack height is  $2^k$ , for which we construct automata  $(\mathcal{A}_i)_{1 \leq i \leq k}$  over a common alphabet  $\Sigma_k$  with the following properties:

1. Each  $\mathcal{A}_i$  has two states  $0_i$  and  $1_i$ , with  $0_i$  the initial state and  $1_i$  the only final one.
2.  $|\Sigma_k| = k$

630 **3.** The intersection of their languages contains a single word of length  $2^k$

631 The construction is simpler if we do this for  $2^k - 1$  instead of  $2^k$ , but this is enough: We can  
 632 introduce a fresh letter  $\#$  and build automata  $\mathcal{A}'_i$  with  $L(\mathcal{A}'_i) = L(\mathcal{A}_i)\#$ .

633 The idea is simply to use  $\Sigma_k = \{\text{inc}_1, \dots, \text{inc}_k\}$ . Each letter is an increment operation  
 634 over a  $k$ -bit binary counter:  $\text{inc}_i$  should be read as “flip the  $i$ th bit from 0 to 1 and all lower  
 635 bits from 1 to 0”. Each automaton  $\mathcal{A}_i$  keeps track of the value of the  $i$ th bit throughout that  
 636 sequence of instructions.

637 Formally,  $\mathcal{A}_i = (\{0_i, 1_i\}, \Sigma_k, \delta_i, 0_i, \{1_i\})$  with  $\delta_i(0_i, \text{inc}_j) = 1_i$  if  $j = i$ ,  $0_i$  if  $j < i$ , and is  
 638 undefined if  $j > i$ . Meanwhile,  $\delta_i(1_i, \text{inc}_j) = 0_i$  if  $j > i$ , and  $1_i$  if  $j \leq i$ . It is easy to check  
 639 that there is a unique word accepted by those automata, corresponding to the only correct  
 640 sequence of instructions to increment a  $k$ -bit binary counter from 0 to  $2^k - 1$ , which clearly  
 641 enforces a single string of length  $2^k - 1$ , as desired.

642 **Step III: Indexed grammars with triply-exponential downward-closure NFA** Suppose  
 643  $\mathcal{A}_1, \dots, \mathcal{A}_k$  are the DFAs over  $\Sigma_k$  built above. Since they will check for exponential stack  
 644 height, but we also need to store the binary digits on the stack, we modify them slightly.  
 645 The automaton  $\mathcal{B}_i$  will work over the alphabet  $\{\perp\} \cup (\Sigma_k \times \{0, 1\})$  and accept exactly the  
 646 words of the form  $(\alpha_1, b_1) \cdots (\alpha_m, b_m) \perp$  where  $\alpha_1 \cdots \alpha_m \in L(\mathcal{A}_i)$ .

647 Note that in order for them to check for exponential stack height Consider the following  
 648 grammar:  $\mathcal{G}_k = (N_k, T, I_k, P_k, S)$  with

- 649 ■  $N_k = \{S, A, B, D, F, Z\}$
- 650 ■  $T = \{a\}$
- 651 ■  $I_k = \{\perp\} \cup (\Sigma_k \times \{0, 1\})$
- 652 ■  $\Delta_k$  contains the following rules:
  - 653 ■  $S \rightarrow Z \perp$
  - 654 ■  $Z \rightarrow Z(\alpha, 0)$  for all  $\alpha \in \Sigma_k$
  - 655 ■  $Z \xrightarrow{B_1, \dots, B_k} D$
  - 656
  - 657 ■  $D \rightarrow AA$
  - 658 ■  $A(\alpha, 1) \rightarrow A$  for all  $\alpha \in \Sigma_k$
  - 659 ■  $A(\alpha, 0) \rightarrow B$  for all  $\alpha \in \Sigma_k$
  - 660 ■  $B \rightarrow Z(\alpha, 1)$  for all  $\alpha \in \Sigma_k$
  - 661 ■  $A \perp \rightarrow F$  and  $F \rightarrow a$

662 The grammar works as follows. Initially, it places  $\perp$  on the stack and switches to  $Z$ . A  
 663 non-terminal  $Z$  will then fill the stack with 0's, which are annotated by  $\alpha \in \Sigma_k$ . After  
 664 pushing these, it non-deterministically stops and then verifies that the stack height is actually  
 665  $2^k$ , by checking that the word consisting of the  $\alpha$ 's is accepted by all the  $\mathcal{A}_i$ 's, using the  
 666 rule  $Z \xrightarrow{B_1, \dots, B_k} D$ . It then splits the  $Z$  into two  $A$ 's, where an increment is performed on  
 667 the number encoded on the stack: It removes the prefix of the form  $1^\ell 0$  and switches to  $Z$ ,  
 668  $B$ . After this, it has to put back  $0^\ell 1$ : To this end, it pushes a single 1 and switches to  $Z$ ,  
 669 which in turn pushes 0's. All this repeats until in each branch, all stack contents encode  
 670 the number  $2^{2^k}$ . When this is achieved, a node  $A[1^{2^k} \perp]$  can be rewritten to  $F$ , and then  
 671  $a$ . Since the nodes are split before each increment (using  $D \rightarrow AA$ ), the final number of  
 672  $a$ -leaves is  $\exp_3(k)$ , deriving  $a^{\exp_3(k)}$ . Moreover, it is straightforward to check that  $a^{\exp_3(k)}$  is  
 673 the only derivable word. A detailed proof can be found in Appendix H.

**Computational hardness** We now use methods from [16] to derive Theorem 3.4 and co-3-NEXP-hardness in Theorem 3.5 from our construction above. For this, we rely on the notion of  $\Delta(f)$  language classes [16], which requires some terminology. A *transducer* is a tuple  $\mathcal{T} = (Q, \Sigma, \Gamma, E, q_0, F)$ , where  $Q$  is a finite set of *states*,  $\Sigma$  is its *input alphabet*,  $\Gamma$  is its *output alphabet*,  $E \subseteq Q \times \Sigma^* \times \Gamma^* \times Q$  is its finite set of *edges*,  $q_0 \in Q$  is its *initial state*, and  $F \subseteq Q$  is its set of *final states*. It describes a relation  $R(\mathcal{T}) \subseteq \Sigma^* \times \Gamma^*$ , namely the set of all pairs  $(u, v)$  for which there are decompositions  $u = u_1 \cdots u_n$  and  $v = v_1 \cdots v_n$ , states  $q_0, q_1, \dots, q_n$ , and edges  $(q_{i-1}, u_i, v_i, q_i) \in E$  for  $i = 1, \dots, n$  with  $q_n \in F$ . For a language  $L \subseteq \Sigma^*$ , we write  $\mathcal{T}(L) = \{v \in \Gamma^* \mid \exists u \in L: (u, v) \in R(\mathcal{T})\}$ .

A *language class* is a class of formal languages, together with some means to represent them, such as grammars or automata. A language class  $\mathcal{C}$  is an *effective full trio* if for a given language  $L$  from  $\mathcal{C}$ , we can effectively compute a description of  $\mathcal{T}(L)$ . Now suppose  $f: \mathbb{N} \rightarrow \mathbb{N}$  is an amplifying function meaning there is a polynomial  $p$  such that  $f(p(n)) \geq f(n)^2$ . Then,  $\mathcal{C}$  is said to be  $\Delta(f)$  if (i) computing  $\mathcal{T}(L)$  can be done in polynomial time and (ii) given  $n$ , one can compute a description of the language  $\{a^{f(n)}\}$  in polynomial time.

In these terms, Proposition 9.1 tells us that the class of indexed languages (represented by indexed grammars) are  $\Delta(\exp_3)$ : Applying rational transductions to indexed languages is well-known to be possible in polynomial time [?].

► **Proposition 9.2.** *The indexed languages (represented by indexed grammars) are  $\Delta(\exp_3)$ .*

We will also need the notion of simple substitutions. For alphabets  $\Sigma, \Gamma$ , a *substitution* is a map  $\sigma: \Sigma \rightarrow 2^{\Gamma^*}$  that replaces each letter in  $\Sigma$  by a language over  $\Gamma$ . For language  $L \subseteq \Sigma^*$ , the language  $\sigma(L)$  is defined in the obvious way. The substitution  $\sigma$  is said to be *simple* for  $L \subseteq \Sigma^*$  if  $\Sigma \subseteq \Gamma$  and there is a letter  $a \in \Sigma$  such that  $\sigma(a') = \{a'\}$  for each  $a' \in \Sigma \setminus \{a\}$ . We say that a language class  $\mathcal{C}$  is *closed under simple substitutions* if for any given  $L$  from  $\mathcal{C}$ , and any simple substitution  $\sigma$  for  $L$ , the language  $\sigma(L)$  belongs to  $\mathcal{C}$ , and a representation can be computed in polynomial time. It is easy to see that the indexed languages are closed under simple substitutions. In [16, Theorem 15], it is shown that downward closure inclusion and equivalence are both  $\text{coTIME}(t)$ -hard for any language class that is  $\Delta(t)$  and closed under simple substitutions. Hence, Proposition 9.1 implies co-3-NEXP-hardness of downward closure inclusion and equivalence.

**DFA size** We can also use the notion of  $\Delta(f)$  language classes to obtain a lower bound on the size of DFAs for downward closures of indexed languages. To this end, we make a slightly more general observation:

► **Proposition 9.3.** *Let  $f: \mathbb{N} \rightarrow \mathbb{N}$  be a function such that a language class  $\mathcal{C}$  is  $\Delta(f)$  and let  $\mathcal{C}$  be closed under simple substitutions. Then there is a family languages  $(L_k)_{k \geq 1}$  with polynomial description sizes such that any DFA for  $L_k \downarrow$  requires at least  $2^{f(k)}$  states.*

**Proof.** We claim that we can construct a representation of

$$L_k = \{uv \mid u, v \in \{0, 1\}^*, |u| = |v| = f(k), u \neq v\}$$

in polynomial time. Note that a DFA for  $L_k \downarrow$  requires at least  $2^{f(k)}$  states: After reading distinct prefixes  $u, u' \in \{0, 1\}^*$  of length  $f(k)$ , the DFA must enter distinct states, as otherwise, it would accept  $uu$ , which does not belong to  $L_k \downarrow$ .

To construct  $L_k$  for given  $k \in \mathbb{N}$ , we begin by building a representation of  $\{a^{f(k)}\}$ . Using a transducer, we then insert a single occurrence of a letter  $\mathbf{b}$  into every word, and then substitute this  $b$  with  $\{\mathbf{b}a^{f(k)}\mathbf{c}\}$ . This yields the language  $\{\mathbf{a}^r \mathbf{b} a^{f(k)} \mathbf{c} a^s \mid r + s = f(k)\}$ .

Using another transducer, we can remove two occurrences of  $a$  within  $\mathbf{a}^r$  and  $\mathbf{a}^s$  to obtain  $\{\mathbf{a}^r \mathbf{b} \mathbf{a}^{f(k)} \mathbf{c} \mathbf{a}^s \mid r + s = f(k) - 2\}$ . A final transducer then replaces (i) each  $\mathbf{a}$  with 0 or 1 and (ii)  $\mathbf{b}$  and  $\mathbf{c}$  by distinct letters in  $\{0, 1\}$ . This results in the language  $L_k$ . ◀

Now, Proposition 9.3 and Proposition 9.2 together directly imply Theorem 3.4.

## 10 Conclusion

We have established tight bounds on the size of an automaton recognizing the downward-closure of an indexed grammar. We rely on an algebraic abstraction of stack contents to translate indexed grammars into context-free ones while preserving the downward-closure. We suspect that this approach can be extended to the whole hierarchy of higher-order pushdown automata.

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## A Normalising indexed grammars

When describing indexed grammars we sometimes use production rules of the form not allowed by our definition of indexed grammar

- (i)  $Af \rightarrow u$  with  $u \in (N \cup T)^* \setminus N$
- (ii)  $A \rightarrow u$  with  $u \in (N \cup T)^* \setminus (N^2 \cup T^*)$

with  $u$  any element of  $(N \cup T)^*$ .

We now formally define how these rules should be eliminated to obtain an indexed grammar as in Definition 2.2.

We start by eliminating rules of the first type: we replace each rule  $Af \rightarrow u$  with  $u \in (N \cup T)^* \setminus N$  by two rules  $Af \rightarrow A'$  and  $A' \rightarrow u$ , with  $A'$  a fresh non-terminal.

It remains to eliminate rules of the form  $A \rightarrow u$  with  $u \in (N \cup T)^* \setminus (N^2 \cup T^*)$ . If  $u = B \in N$  then replace the rule with  $A \rightarrow BC$  and  $C \rightarrow \varepsilon$  with  $C$  a fresh non-terminal. Otherwise, decompose  $u$  as  $u = w_0 A_1 w_1 \dots A_k w_k$  with  $A_1, \dots, A_k \in N$  and  $w_0, \dots, w_k \in T^*$ . Introduce fresh non-terminals  $B_1, \dots, B_k, C_1, \dots, C_{k-1}$ . We replace  $A \rightarrow u$  with rules

- $A \rightarrow W_0 B_1$ ,
- $W_i \rightarrow w_i$  for all  $i \in \{0, \dots, k\}$ ,
- $B_i \rightarrow A_i C_i$  for all  $i \in \{1, \dots, k-1\}$ ,
- $C_i \rightarrow W_i B_{i+1}$  for all  $i \in \{1, \dots, k-1\}$ ,
- $B_k \rightarrow A_k W_k$

Note that  $\sum_{i=0}^k |w_i| \leq |u|$  and  $k \leq |u|$ . Hence, each such rule  $A \rightarrow u$  is replaced by a set of at most  $3|u| + 1$  rules, introducing at most  $2|u| - 1$  non-terminals whose lengths sum up to at most  $8|u| + 6$ . As a consequence, each rule of the form  $Af \rightarrow u$  is replaced by a set of at most  $3|u| + 2$  rules whose lengths sum up to at most  $8|u| + 8$ .

## B Proof of Lemma 5.2

► **Lemma 5.2.** *Let  $\bar{z} = (f_n, X_n) \cdots (f_1, X_1) \in \bar{T}^*$  be a non-empty stack. The following are equivalent:*

1.  $\bar{z}$  is feasible
2.  $\varphi(\bar{z}) \neq \perp$
3. for all  $i > 1$ ,  $X_i = f_i \cdot X_{i-1}$ ,  $\beta(f_{i-1}) \mathcal{R}_{X_i} \alpha(f_i)$ .

**Proof.** From the definitions of  $\mathbb{S}$  and  $\varphi$ , one sees that property 3 is necessary and sufficient to ensure that no product of two consecutive factors in  $\varphi(\bar{z}) = \prod_{i=1}^n \varphi(f_i, X_i)$  is equal to  $\perp$ . It is also clear from the definitions that  $\prod_{i=1}^n \varphi(f_i, X_i) = \perp$  if and only if two consecutive factors multiply to equal  $\perp$ , so we immediately obtain  $2 \Leftrightarrow 3$ .

We now show that  $1 \Rightarrow 3$ . Let  $\bar{z}$  be feasible, so by definition we have a derivation

$$(\alpha(f_1), X_1) \xRightarrow{*}_{\bar{G}} u(\beta(f_n), f_n \cdot X_n)[\bar{z}]v. \quad (*)$$

We proceed by induction on the derivation length. If  $(*)$  has length one, then it must be of the form  $(\alpha(f_1), X_1) \xRightarrow{*}_{\bar{G}} (\beta(f_1), f_1 \cdot X_1)[(f_1, X_1)]$ , whence 3 holds trivially. We now assume that the length is greater than one, and that the first operation is of the form  $(\alpha(f_1), X_1) \rightarrow (B, X_1)(C, X_1)$ . In this case, we may assume without loss of generality that  $(B, X_1)$  derives  $u'(\beta(f_n), f_n \cdot X_n)[\bar{z}]v'$  for some  $u', v' \in \mathbf{SF}$ . Since  $(f_1, X_1)$  must eventually be pushed (and  $(\alpha(f_1), X_1)$  is the only nonterminal which allows this), it follows that

823  $B \mathcal{R}_{X_1} \alpha(f_1)$ , and that there is a derivation  $(\alpha(f_1), X_1) \xRightarrow{*}_{\bar{G}} u''(\beta(f_n), f_n \cdot X_n)[\bar{z}]v''$  for some  
 824  $u'', v'' \in \mathbf{SF}$  that is strictly shorter than  $(*)$ . Hence, 3 holds by induction.

Now let us assume that the length of  $(*)$  is greater than one, and that the first operation is a push (the only remaining possibility). The push operation must be of the form

$$(\alpha(f_1), X_1) \Rightarrow_{\bar{G}} (\beta(f_1), f_1 \cdot X_1)[(f_1, X_1)].$$

825 If  $f_1 \cdot X_1 \neq X_2$ , then it is easy to see that the right hand side cannot derive any term which  
 826 pushes  $(f_2, X_2)$ . Hence, we have

$$827 \quad X_2 = f_1 \cdot X_1 \text{ and } \beta(f_1) \mathcal{R}_{X_2} \alpha(f_2), \quad (**)$$

and there is a derivation  $(\alpha(f_2), X_2)[(f_1, X_1)] \xRightarrow{*}_{\bar{G}} u'(\beta(f_n), f_n \cdot X_n)[\bar{z}]v'$ . Letting  $\bar{z}' = (f_n, X_n) \cdots (f_2, X_2)$ , this implies that  $\bar{z}'$  is feasible with a derivation

$$(\alpha(f_2), X_2) \xRightarrow{*}_{\bar{G}} u'(\beta(f_n), f_n \cdot X_n)[\bar{z}']v'$$

828 that is strictly shorter than  $(*)$ . Hence,  $\bar{z}'$  satisfies 3 by induction, and with  $(**)$  we  
 829 immediately obtain 3 for  $\bar{z}$ .

Finally, we show that  $3 \Rightarrow 1$ . Let us assume that  $\bar{z}$  satisfies 3. Then for  $i = 1, \dots, n-1$  we have

$$(\alpha(f_i), X_i) \Rightarrow_{\bar{G}} (\beta(f_i), X_{i+1})[(f_i, X_i)] \text{ and } (\beta(f_i), X_{i+1}) \xRightarrow{*}_{\bar{G}} u_i(\alpha(f_{i+1}), X_{i+1})v_i$$

830 for some  $u_i, v_i \in \mathbf{SF}$ , as well as  $(\alpha(f_n), X_n) \Rightarrow_{\bar{G}} (\beta(f_n), f_n \cdot X_n)[(f_n, X_n)]$ . It is clear that we  
 831 may combine these derivations to obtain  $(\alpha(f_1), X_1) \xRightarrow{*}_{\bar{G}} u(\beta(f_n), f_n \cdot X_n)[\bar{z}]v$ , proving that  
 832  $\bar{z}$  is feasible.  $\blacktriangleleft$

### 833 **C Proof of Lemma 5.3**

834 **► Lemma 5.3.** *Let  $z \in I^*$ ,  $X \subseteq N$  with  $|z| \geq 1$  and let  $(B, Y, M, A, X) = \varphi(\bar{z}^X)$ , we have,*  
 835 *for all  $C \in X$  and  $D \in Y$ ,*

$$\begin{aligned} 836 & C \mathcal{R}_X A \text{ and } B \mathcal{R}_Y D \\ 837 & \quad \quad \quad \Updownarrow \\ 838 & \text{there exist } u, v \in \mathbf{SF} \text{ such that } (C, X) \xRightarrow{*}_{\bar{G}} u(D, Y)[\bar{z}^X]v. \end{aligned}$$

**Proof.** From the first condition, we have derivations

$$(C, X) \xRightarrow{*}_{\bar{G}} u_C(A, X)v_C \text{ and } (B, X) \xRightarrow{*}_{\bar{G}} u_D(D, X)v_D$$

for some  $u_C, v_C, u_D, v_D \in \mathbf{SF}$ . Let  $z = f_n \cdots f_1$ . From the definition of  $\varphi$  (and the fact that  $\varphi(\bar{z}^X) \neq \perp$ ) it follows easily that  $A = \alpha(f_1)$ ,  $B = \beta(f_n)$  and  $Y = z \cdot X$ . Hence, Lemma 5.2 ensures that there is a derivation

$$(A, X) \xRightarrow{*}_{\bar{G}} u'(B, Y)[\bar{z}^X]v'$$

for some  $u', v' \in \mathbf{SF}$ , which we combine with the derivations above to obtain

$$(C, X) \xRightarrow{*}_{\bar{G}} u(D, Y)[\bar{z}^X]v,$$

839 proving one direction.

840 For the other implication, suppose such a derivation exists. Eventually,  $(f_1, X)$  must be  
 841 pushed onto an empty stack, and since  $(A, X)$  is the only non-terminal which facilitates  
 842 this operation, it follows that  $C \mathcal{R}_X A$ . Similarly, the derivation must eventually push the  
 843 topmost symbol in  $\bar{z}^X$ , and the only non-terminal which can result from this operation  
 844 is  $(B, Y)$ . This implies that  $(B, Y)[\bar{z}^X] \xRightarrow{*}_{\bar{G}} u(D, Y)[\bar{z}^X]v$ , hence  $B \mathcal{R}_Y D$ , completing the  
 845 proof.  $\blacktriangleleft$



## D Proof of Lemma 5.4

► **Lemma 5.4.** Let  $z \in I^*$ ,  $X \subseteq N$  with  $|z| \geq 1$  and let  $(B, Y, M, A, X) = \varphi(\bar{z}^X)$ , we have

$$M(D, C) = \top$$

$$\Downarrow$$

$C \in X$ ,  $D \in Y$  and there exist  $u, v \in (X \cup T)^*$  such that  $D[z] \xRightarrow{*} guCv$ .

**Proof.** We proceed by induction on  $z$ . If  $|z| = 1$  then we have  $\bar{z}^X = (f, X)$  with  $\varphi(f, X) = (B, Y, M, A, X)$ . Then the equivalence holds simply by definition.

If  $|z| > 1$  then let  $z = fz'$ . Let  $Z = z' \cdot X$ , we have  $\bar{z}^X = (f, Z)\bar{z}'^X$ . Define  $(B_f, Y, M_f, A_f, Z) = \varphi(f, Z)$  and  $(B', Z, M', A', X) = \varphi(\bar{z}'^X)$ . Note that  $M = M_f M'$ . We prove the two directions separately.

⇒ Suppose  $M(D, C) = \top$ . Then there exists  $E$  such that  $\mu_f(D, E) = \mu'(E, C) = \top$ .

By induction hypothesis applied to  $f$  we have:  $D \in Y$ ,  $E \in Z$  and there exist  $u_1, v_1 \in (Z \cup T)^*$  such that  $D[f] \xRightarrow{*} gu_1 E v_1$ .

Furthermore, by induction hypothesis applied to  $z'$  we have  $C \in X$  and there exist  $u_2, v_2 \in (X \cup T)^*$  such that  $E[z'] \xRightarrow{*} gu_2 C v_2$ .

Since  $Z = z' \cdot X$ , there exist  $u_3, v_3 \in (X \cup T)^*$  such that  $u_1^{z'} \xRightarrow{*} gu_3$  and  $v_1^{z'} \xRightarrow{*} gv_3$ . Here  $u_1^{z'}$  and  $v_1^{z'}$  denote the sentential forms  $u_1$  and  $v_1$  where  $z'$  has been pushed on top of the stack of every non-terminal.

Combining those facts, we get  $D[z] \xRightarrow{*} gu_1^{z'} E[z'] v_1^{z'} \xRightarrow{*} gu_3 E[z'] v_3 \xRightarrow{*} gu_3 u_2 C v_2 v_3$ .

⇐ Suppose  $D \in Y$ ,  $C \in X$  and there exist  $u, v \in (X \cup T)^*$  such that  $D[z] \xRightarrow{*} guCv$ .

Then we have  $D[f] \xRightarrow{*} gu_1 E v_1$ ,  $E[z'] \xRightarrow{*} gu_2 C v_2$ ,  $u_1^{z'} \xRightarrow{*} gu_3$  and  $v_1^{z'} \xRightarrow{*} gv_3$  for some  $u_1, u_2, u_3, v_1, v_2, v_3$  such that  $u = u_3 u_2$  and  $v = v_2 v_3$ .

notation  $u^z$

As a consequence, we have  $u_2, u_3, v_2, v_3 \in (X \cup T)^*$ . Since  $z' \cdot X = Z$ , we get that  $u_1, v_1 \in (Z \cup T)^*$ . Since  $C \in X$ , the derivation  $E[z'] \xRightarrow{*} gu_2 C v_2 \in (X \cup T)^*$  proves that  $E \in z' \cdot X = Z$ .

By induction hypothesis, we have  $M'(D, E) = M_f(E, C) = \top$  and thus  $M(D, C) = \top$ .

## E Proof of Lemma 7.5

In this section we prove the following statement.

► **Lemma 7.5.** For all  $z \in \bar{I}^*$ ,  $\text{push}(z \blacktriangleright \varepsilon)$  is of size at most exponential in  $N$ .

Its proof is very similar to the one of Theorem 26 in [8].

Let us start by recalling some facts on Green relations.

► **Lemma E.1** ([13]). In a finite semigroup  $\mathbf{S}$ , every  $\mathcal{H}$ -class contains at most one idempotent.

Find the right chapter in [13].

► **Lemma E.2** ([13]). In a finite semigroup  $\mathbf{S}$ , for all  $x, y \in \mathbf{S}$ , if  $x \mathcal{J} y$  and  $x \leq_{\mathcal{L}} y$  (resp.  $x \leq_{\mathcal{R}} y$ ) then  $x \mathcal{L} y$  (resp.  $x \mathcal{R} y$ ).

Define the *Ramsey function* of  $\mathbf{S}$  as follows: for all  $k \in \mathbb{N}$ ,  $\mathcal{R}_{\mathbf{S}}(k)$  is the minimal  $n$  such that every word of length  $n$  contains a factor  $u$  such that  $\psi_{\mathbf{S}}(u)$  is an idempotent.

885 ► **Theorem E.3** ([10], Theorem 1). *For all finite semigroup<sup>2</sup>  $\mathbf{S}$ , for all  $k \in \mathbb{N}$ ,*

$$886 \quad \mathcal{R}_{\mathbf{S}}(k) \leq (k|\mathbf{S}|^4)^{JL(\mathbf{S})}.$$

887 ► **Theorem E.4** ([10], Theorem 2). *The regular  $\mathcal{J}$ -length of  $\mathbb{M}$  is  $JL(\mathbb{M}) = \frac{N^2+N+2}{2}$ .*

888 ► **Theorem E.5.** *The regular  $\mathcal{J}$ -length of  $\mathbb{S}$  is at most  $\frac{(N^2+N+2)}{2} + 1$ .*

889 **Proof.** Let us start by using another definition of the regular  $\mathcal{J}$ -length. By [10, Theorem  
890 ???], the regular  $\mathcal{J}$ -length of  $\mathbb{S}$  is the largest  $m$  such that there is an injective homomorphism  
891 from the max monoid  $(\{1, \dots, m\}, \max, m)$  to  $\mathbb{S}$ .

892 Let  $m \in \mathbb{N}$ , let  $\theta : \{1, \dots, m\} \rightarrow \mathbb{S}$  be such a homomorphism. We must show that  
893  $m \leq \frac{(N^2+N+2)}{2} + 1$ .

894 If  $\perp$  is in the image of  $\theta$  then  $\theta(m) = \perp$ . Since all  $i$  in  $\{1, \dots, m-1\}$  are idempotents,  
895 the image of each  $i$  by  $\theta$  must also be an idempotent. As a result, we can set  $\theta(i) =$   
896  $(B_i, X_i, M_i, A_i, X_i)$  for all  $i < m$ , with  $M_i^2 = M_i$ .

897 Furthermore, for all  $i < j < m$ , since  $\max(i, j) = j$ , we must have  $(B_j, Y_j, M_j, A_j, X_j) \cdot$   
898  $(B_i, Y_i, M_i, A_i, X_i) = (B_i, Y_i, M_i, A_i, X_i) \cdot (B_j, Y_j, M_j, A_j, X_j) = (B_j, Y_j, M_j, A_j, X_j)$ . We  
899 infer that  $X_j = X_i$ ,  $B_j = B_i$  and  $A_j = A_i$  for all  $i < j$ . Since  $\theta$  is injective,  $M_1, \dots, M_{m-1}$   
900 must be distinct.

901 Define the function  $\theta' : \{1, \dots, m-1\}$  mapping each  $i$  to  $M'_i$ . It suffices to observe  
902 that  $\theta_p$  is an injective homomorphism from the max monoid of size  $m-1$  to  $\mathbb{M}$ . As a  
903 consequence, we have  $m-1 \leq JL(\mathbb{M})$ . By Theorem E.4, we have  $JL(\mathbb{M}) \leq \frac{N^2+N+2}{2}$ . As a  
904 result,  $m \leq \frac{(N^2+N+2)}{2} + 1$ . ◀

905 Observe that the size of  $\mathbb{S}$  is bounded by  $N^2 2^{4N^2+8N}$ . Define  $K = ((N+1)N^2 2^{4N^2+8N})^{\frac{(N^2+N+2)}{2}+1}$ .  
906 By combining the results above, we obtain the following corollary.

907 ► **Corollary E.6.** *Let  $z \in \mathbb{S}^*$  with  $|z| \geq K$ , there exist  $v_0, \dots, v_N \in \mathbb{S}^*$  and  $e \in \text{Idem}(\mathbb{S})$  such  
908 that*

909 ■  $v_0 \dots v_N$  is a factor of  $z$

910 ■  $\varphi(v_0) = \dots = \varphi(v_N) = e$

911 In what follows we distinguish the size of an  $h$ -summary/ $h$ -block from its **length**, which is  
912 simply its length as a word of  $h$ -atoms, plus one (for the  $(h-1)$ -summary at the start).

913 ► **Lemma E.7.** *For all  $z \in \bar{I}^*$ ,  $\text{push}(z \blacktriangleright \varepsilon) = \sigma' u B_1 \dots B_K$  is such that  $u$  and all  $B_i$  have  
914 length at most  $K$ .*

infix or factor

915 **Proof.** If  $\text{push}(z \blacktriangleright \varepsilon) = \sigma' u B_1 \dots B_k$  then  $u$  cannot have an infix of the form  $v_0 \dots v_N$  with  
916  $\varphi(v_i) = e$  for all  $i$ , for any  $e \in \text{Idem}(\mathbb{S})$ . This is because when such a pattern appears  $u$  is  
917 turned into an  $h$ -block.

918 As a consequence, by Corollary E.6,  $u$  has length at most  $K-1$ .

919 Therefore, since all blocks are created by concatenating an  $h$ -atom with such a  $u$ , they  
920 have **length** at most  $K$ . ◀

921 ► **Lemma E.8.** *For all  $z \in \bar{I}^*$ ,  $\text{push}(z \blacktriangleright \varepsilon)$  has at most  $|\mathbb{S}|^{4JL(\mathbb{S})+1}$  blocks.*

<sup>2</sup> The theorem is stated for finite monoids in [10], but it is straightforward to adapt it to finite semigroups.

**Proof.** Let  $\sigma = \sigma' u B_1 \dots B_m$  be an  $h$ -summary. For each  $i$  let  $B_i = e_i^+ v_{i,1}, \dots, v_{i,N} w_i$  and for each  $i < j$  define  $\alpha_i = \psi_{\mathbb{S}}(v_{i,1} \dots v_{i,N} w_i B_{i+1} \dots B_{j-1} e_j)$ .

Suppose by contradiction that  $m > |\mathbb{S}|^{4JL(\mathbb{S})+1}$ , then by pigeonhole principle there exist  $e \in \text{Idem}(\mathbb{M})$  and  $i_1 < \dots < i_p \in \{1, \dots, K\}$  such that  $p > (|\mathbb{S}|)^{4JL(\mathbb{S})}$  and  $e_{i_k} = e$  for all  $k$ .

Then by Theorem E.3, there exist  $k, \ell$  such that  $\psi_{\mathbb{S}}(\alpha_{i_k} \dots \alpha_{i_\ell})$  is an idempotent  $e'$ .

Since  $\sigma$  is an  $h$ -summary, we have  $\text{height}(e') = h = \text{height}(\varphi_{\mathbb{M}}(\alpha_{i_1} \dots \alpha_{i_p})) = \text{height}(e)$ .

Since  $e' \leq_{\mathcal{J}} e$ , we have  $e \mathcal{J} e'$ . Furthermore, since  $e' \leq_{\mathcal{R}} \alpha_{i_k} \leq_{\mathcal{R}} \varphi_{\mathbb{M}}(\pi(v_{i_k,1})) = e$ , we have  $e' \leq_{\mathcal{R}} e$  and thus  $e \mathcal{R} e'$  by Lemma E.1. Similarly, since  $e' \leq_{\mathcal{L}} \alpha_{i_\ell} \leq_{\mathcal{L}} e$  we have  $e' \leq_{\mathcal{L}} e$  and thus  $e \mathcal{L} e'$ .

We obtain  $e \mathcal{H} e'$ . By Lemma E.2, this implies  $e = e'$ . This is a contradiction since then the blocks from  $B_{i_k}$  to  $B_{i_\ell}$  should have been merged when  $B_{i_k}$  was created. As a result, we must have  $m \leq |\mathbb{S}|^{4JL(\mathbb{S})+1}$ .  $\blacktriangleleft$

We now have all necessary tools to show Lemma 7.5.

**Proof of Lemma 7.5.** We show that for all  $z$  and  $h$ , if  $\text{push}(z \blacktriangleright \varepsilon)$  is an  $h$ -summary then it has size at most  $((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1)^h$ .

We do an induction on  $h$ . The property trivially holds for  $h = 0$ . Let  $h > 0$ , suppose the property holds for  $h - 1$ .

Let  $\text{push}(z \blacktriangleright \varepsilon) = \sigma' u B_1 \dots B_m$ . By Lemma E.7 we have  $m \leq |\mathbb{S}|^{4JL(\mathbb{S})+1}$ . By Lemma E.8 we have  $|u| \leq K$  and  $|B_i| \leq K$  for all  $i$ .

Therefore  $\text{push}(z \blacktriangleright \varepsilon)$  has length at most  $(|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1$ .

Let  $a\sigma''$  be an  $h$ -atom appearing in  $z$ , with  $\sigma''$  an  $h - 1$ -summary. Observe that there must be a strict infix  $z''$  of  $z$  such that  $\text{push}(z'' \blacktriangleright \varepsilon) = \sigma''$ . Hence by induction hypothesis  $\sigma''$  has size at most  $((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1)^{h-1}$ . Thus every  $h$ -atom has size at most  $((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1)^{h-1} + 1 \leq 2((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1)^{h-1}$ . With the bound on its length, we conclude that  $\text{push}(z \blacktriangleright \varepsilon)$  has size at most

$$2((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1)^{h-1}((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1) \leq ((|\mathbb{S}|^{4JL(\mathbb{S})+1} + 1)K + 1)^h.$$

We obtain the result by applying this bound with  $h = JL(\mathbb{S})$ , which is at most  $\frac{(N^2+N+2)}{2} + 1$  by Theorem E.5.  $\blacktriangleleft$

## F Proof of Lemma 8.4

**Lemma 8.4.** Let  $(A, X, \sigma_1) \rightarrow (B, Y, \sigma_2)$  be a rule of  $\mathcal{C}_G$  with  $\sigma_2 \in \text{pop}((f, Y) \blacktriangleleft \sigma_1)$ . Let  $\bar{z}_1$  an *expansion* of  $\sigma_1$ . Then there exists  $\bar{z}_2$  an *expansion* of  $\sigma_2$  such that

$$(A, X)[\bar{z}_1](\rightarrow_{\text{pump}} + \rightarrow_{\text{skip}})^*(A, X)[(f, Y)\bar{z}_2].$$

**Proof.** We prove this statement by induction on the *height* of  $\sigma_1$ . If  $\sigma_1$  has *height* 0 then it is  $\varepsilon$  and  $\text{pop}((f, Y) \blacktriangleleft \sigma_1)$  is empty.

If  $\sigma_1$  has *height*  $h > 0$  then we distinguish cases according to its shape. In most cases we will simply show that  $\bar{z}_1$  is of the form  $(f, Y)\bar{z}_2$  with  $\bar{z}_2$  a  $K$ -expansion of  $\sigma_2$ . Let  $\sigma_2 = \sigma' u B_1 \dots B_k$ .

1. If  $\text{height}((f, Y)\sigma_2) > h$  then  $\sigma_1 = (f, Y)\sigma_2$ . Then by definition  $\bar{z}_1$  must be of the form  $(f, Y)\bar{z}_2$  with  $\bar{z}_2$  a  $K$ -expansion of  $\sigma_2$  (hence also a  $K/D$ -expansion by Remark ??).
2. Otherwise, we have  $\text{height}((f, Y)\sigma_2) = h$

- a. if  $\text{height}((f, Y)\sigma') < h$  then  $\sigma_1 = (\text{push}((f, Y) \blacktriangleright \sigma'))uB_1 \dots B_k$ . Then we have  $\bar{z}_1 = z''z^u z_1 \dots z_k$  with  $z''$  a  $K$ -expansion of  $\text{push}((f, Y) \blacktriangleright \sigma')$ ,  $z^u$  of  $u$  and  $z_i$  of  $B_i$  for all  $i$ . By construction of  $\mathcal{C}_G$ ,  $(A, X, \text{push}((f, Y) \blacktriangleright \sigma')) \rightarrow (B, Y, \sigma')$  must be a rule

of  $\mathcal{C}_G$ . Hence, by induction hypothesis,  $(A, X)[z''] \rightarrow_{skip}^* (A, X)[(f, Y)z']$  with  $z'$  a  $K/D$ -expansion of  $\sigma'$ . As a result, we have  $(A, X)[\bar{z}_1] = (A, X)[z''z^u z_1 \dots z_k] \rightarrow_{skip}^* (A, X)[(f, Y)z'z^u z_1 \dots z_k]$ . By Remark ??, since  $z^u z_1 \dots z_k$  is a  $K$ -expansion of  $uB_1 \dots B_k$ , it is also a  $K/D$ -expansion of it, and thus  $z'z^u z_1 \dots z_k$  is a  $K/D$ -expansion of  $\sigma'uB_1 \dots B_k = \sigma_2$ .

b. Otherwise,  $\text{height}((f, Y)\sigma') = h$  and  $(f, Y)\sigma'$  is an  $h$ -atom.

i. If  $((f, Y)\sigma')u$  is of the form  $u_1 \dots u_N v_1 \dots v_N w$  with  $\varphi(u_i) = \varphi(v_i) = e$  for all  $i$ , for some  $e \in \text{Idem}(\mathbb{S})$ , we have two cases.

A. If there exists  $j$  such that  $B_j$  is of the form  $u'_1 \dots u'_N e^+ v'_1 \dots v'_N w'$  and

$\varphi(v_1 \dots v_N B_1 \dots B_{j-1} u'_1 \dots u'_N) = e$  then we have

$\sigma_1 = (u_1 \dots u_N e^+ v'_1 \dots v'_N w') B_{j+1} \dots B_k$ . In that case,  $\bar{z}_1 = z_1^u \dots z_N^u z_e' z_1^{v'} \dots z_N^{v'} z^{w'} z_{j+1} \dots z_k$  where  $z_j^u$  and  $z_j^{v'}$  are  $K$ -expansions of  $u_j$  and  $v'_j$  for all  $j$ ,  $z_e'$  is a  $K$ -expansion of  $e^+$  and  $z^{w'}$  of  $w'$ .

Let  $z_e$  be a  $K/D$ -expansion of  $v_1 \dots v_N B_1 \dots B_{j-1} u'_1 \dots u'_N$  of length at most  $K$ . It exists by Lemma ?. Since  $\varphi(v_1 \dots v_N B_1 \dots B_{j-1} u'_1 \dots u'_N) = e$ ,  $z_e$  is a realisation of  $e$ , of length at most  $K$ . As a consequence, we have  $z_e' = z_- z_e z_+$  with  $\varphi(z_-) = \varphi(z_+) = e$ .

We can thus use a pump rule and a skip rule:

$$\begin{aligned} (A, X)[\bar{z}_1] &= (A, X)[z_1^u \dots z_N^u z_- z_e z_+ z_1^{v'} \dots z_N^{v'} z^{w'} z_{j+1} \dots z_k] \\ &\rightarrow_{pump} (A, X)[z_1^u \dots z_N^u z_e z_1^{v'} \dots z_N^{v'} z_1^u \dots z_N^u z_- z_e z_+ z_1^{v'} \dots z_N^{v'} z^{w'} z_{j+1} \dots z_k] \\ &\rightarrow_{skip} (A, X)[z_1^u \dots z_N^u z_e z_1^{v'} \dots z_N^{v'} z^{w'} z_{j+1} \dots z_k] \end{aligned}$$

Since  $z_e$  is a  $K/D$ -expansion of  $v_1 \dots v_N B_1 \dots B_{j-1} u'_1 \dots u'_N$ ,  $z_1^u \dots z_N^u z_e z_1^{v'} \dots z_N^{v'} z^{w'} z_{j+1} \dots z_k$  is a  $K/D$ -expansion of  $u_1 \dots u_N v_1 \dots v_N B_1 \dots B_{j-1} u'_1 \dots u'_N v'_1 \dots v'_N w' B_{j+1} \dots B_k = ((f, Y)\sigma')uB_1 \dots B_{j-1} u'_1 \dots u'_N v'_1 \dots v'_N w' B_{j+1} \dots B_k = (f, Y)\sigma_2$ .

B. Otherwise  $\sigma_1 = (u_1 \dots u_N e^+ v_1 \dots v_N) B_1 \dots B_k$  with  $B = u_1 \dots u_N e^+ v_1 \dots v_N$ .

Let  $z_e'$  be a  $K$ -expansion of  $e^+$ . Observe that since  $\varphi(u_1 \dots u_N) = e$ ,  $\varphi(z_1^u \dots z_N^u) = e$ . Further,  $\varphi(z_e') = e$ , and thus we have  $(B, Y)[(f, X)\bar{z}_1] = (B, Y)[z_1^u \dots z_N^u z_1^{v'} \dots z_N^{v'} z_1 \dots z_k] \rightarrow_{pump} (B, Y)[z_1^u \dots z_N^u z_e' z_1^u \dots z_N^u z_1^{v'} \dots z_N^{v'} z_1 \dots z_k]$ . Since  $z_e'$  is a  $K$ -expansion of  $e^+$ , so is  $z_e' z_1^u \dots z_N^u$ . As a consequence, the resulting stack content  $z_1^u \dots z_N^u z_e' z_1^u \dots z_N^u z_1^{v'} \dots z_N^{v'} z_1 \dots z_k$  is a  $K$ -expansion of  $\sigma_2$ .

ii. Otherwise, we have  $\sigma_1 = ((f, Y)\sigma')uB_1 \dots B_k$  and thus  $\bar{z}_1 = (f, Y)\bar{z}_2$  with  $\bar{z}_2$  a  $K$ -expansion of  $\sigma_2$ .

## G Proof of Lemma 6.2

We prove the following lemma.

► **Lemma 6.2.**  $L^{pump, skip}(\bar{\mathcal{G}}) \subseteq L(\bar{\mathcal{G}}) \downarrow$

► **Lemma G.1.** Let  $(A, X)[\bar{z}]$  be a term of  $\bar{\mathcal{G}}$ ,  $e \in \text{Idem}(\mathbb{S})$  and  $z_e \in \bar{I}^*$  such that  $\varphi(z_e) = e$ . Then  $L((A, X)[z_e \bar{z}]) \subseteq L((A, X)[\bar{z}]) \downarrow$ .

**Proof.** Let  $w \in L((A, X)[\bar{z}])$ . There is a derivation  $(A, X)[\bar{z}] \xrightarrow{*} \bar{\mathcal{G}} w$ . Since  $\varphi(z_e) = e$ , by Lemma 5.2,  $z_e$  is feasible, i.e., there is a derivation  $(A, X) \xrightarrow{*} \bar{\mathcal{G}} u(B, X)[z_e]v$  with  $u, v \in \mathbf{SF}$ .

Furthermore, since  $e$  is idempotent, by definition of the product of  $\mathbb{S}$ , we have  $B \mathcal{R}_X A$ , i.e., there is a derivation  $(B, X) \xRightarrow{*}_{\bar{\mathcal{G}}} u'(A, X)v'$ .

In total, we get  $(A, X)[\bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} u(B, X)[z_e]v \xRightarrow{*}_{\bar{\mathcal{G}}} uu'(A, X)[z_e\bar{z}]v'v \xRightarrow{*}_{\bar{\mathcal{G}}} uu'wv'v$ . By Lemma ??,  $w \in L((A, X)[\bar{z}]) \downarrow$   $\blacktriangleleft$

► **Lemma G.2.** Let  $(A, X)[v_1 \dots v_N z_e v_1 \dots v_N \bar{z}]$  be a term of  $\bar{\mathcal{G}}$ , and  $e = (B, X, M, A, X) \in \text{Idem}(\mathbb{S})$  such that  $z_e \in T^*$  such that  $\varphi(v_1) = \dots = \varphi(v_N) = \varphi(z_e)$ . Then  $L((A, X)[v_1 \dots v_N \bar{z}]) \subseteq L((A, X)[v_1 \dots v_N z_e v_1 \dots v_N \bar{z}]) \downarrow$ .

**Proof.** Let  $w \in L((A, X)[u_1 \dots u_N \bar{z}])$ . Let  $\tau$  be a derivation tree for  $(A, X)[u_1 \dots u_N \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w$ . Consider  $V$  the set of nodes  $\nu$  such that:

1. Either  $\nu$  is a leaf with a label in  $T^*$  and  $\bar{z}$  is a suffix of the stack content of all its strict ancestors,
2. or  $\nu$  is labeled  $(C, Y)[v_i \dots v_N \bar{z}]$  for some  $i \in \{1, \dots, N+1\}$  and  $v_i \dots v_N \bar{z}$  is a suffix of the stack content of all its ancestors. If  $M(B, B) = \top$  we say that  $\nu$  is a *special node*.

► **Claim G.3.** Every branch of  $\tau$  contains either a node of the first type or a special node.

**Proof.** Consider a branch of  $\tau$ . If the leaf of this branch is not of the first type, then the prefix  $v_1 \dots v_N$  must be fully popped.

Therefore, there are nodes  $\nu_1, \dots, \nu_{N+1}$  of the second type along this branch: each  $\nu_i$  is labeled  $(A_i, X_i)[v_i \dots v_N \bar{z}]$  and all its ancestors have a stack content with suffix  $v_i \dots v_N \bar{z}$ .

In consequence, for all  $i$  we can define the derivation tree obtained by restricting  $\tau$  to descendants of  $\nu_i$ , and then removing all nodes where  $v_{i+1} \dots v_N \bar{z}$  is not a suffix of the stack and their descendants, as well as strict descendants of  $\nu_{i+1}$ . By definition of  $\nu_{i+1}$ , it is a leaf of the resulting tree.

We obtain a derivation  $(A_i, X_i)[v_i \dots v_N \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w_i(A_{i+1}, X_{i+1})[v_{i+1} \dots v_N \bar{z}]w'_i$  with  $w_i, w'_i \in \mathbf{SF}$ . By definition of  $M$ , since  $\varphi(v_i) = e$ , we have  $X_i = X_{i+1} = X$  and  $M(A_i, A_{i+1}) = \top$ .

By Lemma ??, since  $M(A_i, A_{i+1}) = \top$  for all  $i$ , there exists  $i$  such that  $M(A_i, A_i) = \top$ . As a consequence,  $\nu_i$  is a special node.  $\blacktriangleleft$

Let  $\nu$  be a special node, labeled  $(C, X)[u_i \dots u_N \bar{z}]$  with  $M(C, C) = \top$ . Since  $\varphi(v_1) = \dots = \varphi(v_N) = \varphi(z_e) = e$ ,  $\varphi(v_i \dots v_N z_e v_1 \dots v_{i-1}) = e$ . As a consequence, since  $M(C, C) = \top$ , we have a derivation  $(C, X)[v_i \dots v_N z_e v_1 \dots v_{i-1}] \xRightarrow{*}_{\bar{\mathcal{G}}} w_-(C, X)w_+$  with  $w_-, w_+ \in \mathbf{SF}$ .

Consider the set of minimal nodes for the ancestor relation, which are either of the first type or special. They form a maximal antichain for the ancestor relation, and intersect every branch.

We can thus define the subtree whose root is the same as  $\tau$  and whose leaves are those nodes. By definition, every node in this tree is labeled with a term whose stack has  $\bar{z}$  as suffix. Define  $\tau'$  the tree obtained by replacing each suffix  $\bar{z}$  with  $z_e u_1 \dots u_N \bar{z}$ . The resulting tree  $\tau'$  is still a valid derivation tree.

Below each leaf  $\nu$  of  $\tau'$  which is a special node we append with a label of the form  $(C, X)[v_i \dots v_N \bar{z}]$ , we append a derivation tree for

$$(C, X)[v_i \dots v_N z_e v_1 \dots v_N \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w_-(C, X)[v_i \dots v_N \bar{z}]w_+ \text{ with } w_-, w_+ \in \mathbf{SF}$$

. Then we append the subtree of  $\tau$  rooted in  $\nu$  at the resulting leaf labeled  $(C, X)[v_i \dots v_N \bar{z}]$ .

We obtain a derivation tree from  $(A, X)[z'v_1 \dots v_N]$  to a sentential form  $\bar{w}$  of which  $w$  is a subword. Since  $\bar{\mathcal{G}}$  is sound and  $(A, X)[z'v_1 \dots v_N]$  is reachable, there is a word  $w' \in T^*$  such that  $\bar{w} \xRightarrow{*}_{\bar{\mathcal{G}}} w'$ . Since  $w \preceq \bar{w}$ ,  $w \preceq w'$ , proving the result.  $\blacktriangleleft$

1049 **Proof of Lemma 6.2.** We show that for all **reachable** sentential form  $u$  of  $\bar{\mathcal{G}}$ ,  $L(u)^{pump,skip} \subseteq$   
 1050  $L(u) \downarrow$ .

1051 Let  $w \in L(u)^{pump,skip}$ , we proceed by induction on the derivation  $u \xRightarrow{*}_{\bar{\mathcal{G}}}^{pump,skip} w$ , and  
 1052 distinguish cases according to the first step.

- 1053 ■ If the first step is  $u \Rightarrow_{\bar{\mathcal{G}}} u'$  then  $u' \xRightarrow{*}_{\bar{\mathcal{G}}}^{pump,skip} w$ . By induction hypothesis,  $u' \Rightarrow_{\bar{\mathcal{G}}} w'$  with  
 1054  $w \preceq w'$ , hence  $u \Rightarrow_{\bar{\mathcal{G}}} u' \xRightarrow{*}_{\bar{\mathcal{G}}} w'$ .
- 1055 ■ If the first step is  $u = u_1(A, X)[z_e \bar{z}]u_2 \rightarrow_{pump} u_1(A, X)[z'_e \bar{z}]u_2 = u'$  then  $u' \xRightarrow{*}_{\bar{\mathcal{G}}}^{pump,skip} w$ .  
 1056 By induction hypothesis,  $u' \Rightarrow_{\bar{\mathcal{G}}} w'$  with  $w \preceq w'$ . Let  $w' = w_1 w_A w_2$  such that  $u_1 \xRightarrow{*}_{\bar{\mathcal{G}}} w_1$ ,  
 1057  $u_2 \xRightarrow{*}_{\bar{\mathcal{G}}} w_2$  and  $(A, X)[z'_e \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w_A$ . By Lemma G.1,  $u \xRightarrow{*}_{\bar{\mathcal{G}}} w_1 w'_A w_2$  with  $w_A \preceq w'_A$ . We  
 1058 obtain the result since  $w \preceq w' \preceq w_1 w'_A w_2$ .
- 1059 ■ If the first step is  $u = u_-(A, X)[z_e v_1 \dots v_N \bar{z}]u_+ \rightarrow_{skip} u_-(A, X)[v_1 \dots v_N \bar{z}]u_+ = u'$   
 1060 then  $u' \xRightarrow{*}_{\bar{\mathcal{G}}}^{pump,skip} w$ . By induction hypothesis,  $u' \Rightarrow_{\bar{\mathcal{G}}} w'$  with  $w \preceq w'$ . Let  $w' =$   
 1061  $w_- w_A w_+$  such that  $u_- \xRightarrow{*}_{\bar{\mathcal{G}}} w_-$ ,  $u_+ \xRightarrow{*}_{\bar{\mathcal{G}}} w_+$  and  $(A, X)[v_1 \dots v_N \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w_A$ . By Lemma G.1,  
 1062  $(A, X)[v_1 \dots v_N z_e v_1 \dots v_N \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w'_A$  with  $w_A \preceq w'_A$ . By Lemma G.2,  $(A, X)[z_e v_1 \dots v_N \bar{z}] \xRightarrow{*}_{\bar{\mathcal{G}}} w''_A$   
 1063 with  $w'_A \preceq w''_A$ .  
 1064 We thus have  $u \xRightarrow{*}_{\bar{\mathcal{G}}} w_- w''_A w_+$ . Further,  $u_- w''_A w_+ \xRightarrow{*}_{\bar{\mathcal{G}}} w_- w'_A w_+$ . We obtain the result since  
 1065  $w \preceq w' \preceq w_- w'_A w_+ \preceq w_- w''_A w_+$ .

1066 The induction is proven. To obtain the lemma, it suffices to apply it to  $u = (S, \mathbf{U})$ . ◀

## 1067 **H** Additional material from Section 9

1068 ► **Lemma H.1.** *There is an execution of  $\mathcal{G}_k$  that produces the word  $\mathbf{a}^{\exp_3(k)}$*

1069 **Proof.** Let  $\alpha_1 \dots \alpha_m \in \Sigma_k^*$  be the unique word accepted by all  $\mathcal{A}_i$ , with  $m = 2^k$ . For all  
 1070  $b_1, \dots, b_m \in \{0, 1\}$ , we write  $\overline{b_1 \dots b_m}$  for the number in  $[0, 2^m - 1]$  whose binary representation  
 1071 over  $m$  bits is  $b_1 \dots b_m$ ,  $b_1$  being the bit of smallest weight.

1072 We show that for all  $b_1, \dots, b_m \in \{0, 1\}$ , if  $M = \overline{b_1 \dots b_m}$  then  $Z(\alpha_1, b_1) \dots (\alpha_m, b_m) \perp$   
 1073 produces  $\mathbf{a}^{2^{2^m - M}}$ , by induction on  $2^m - M$ .

1074 We can apply  $Z \xrightarrow{B_1, \dots, B_k} D$  and  $D \rightarrow AA$  to obtain two copies of  $A(\alpha_1, b_1) \dots (\alpha_m, b_m) f_{\perp}$ .  
 1075 This is because  $\alpha_1 \dots \alpha_m$  is accepted by all  $\mathcal{A}_i$ . We are left with

$$1076 \quad A[(\alpha_1, b_1) \dots (\alpha_m, b_m) \perp] A[(\alpha_1, b_1) \dots (\alpha_m, b_m) \perp].$$

1077 If  $b_i = 1$  for all  $i$ , then we can apply  $A(\alpha_i, 1) \rightarrow A$  for each  $i$  and then  $A \perp \rightarrow F$  and  
 1078  $F \rightarrow \mathbf{a}$  to obtain  $\mathbf{aa}$ , which is what we want since we would then have  $M = 2^m - 1$  and thus  
 1079  $\mathbf{a}^{2^{2^m - M}} = \mathbf{a}^2$ .

1080 Otherwise, let  $j$  be the least index such that  $b_j = 0$ . It suffices to show that  $A(\alpha_1, b_1) \dots (\alpha_m, b_m) \perp$   
 1081 produces  $\mathbf{a}^{2^{2^m - M - 1}}$ . We have  $M + 1 = \overline{1^{j-1} 0 b_{j+1} \dots b_m} + 1 = \overline{0^{j-1} 1 b_{j+1} \dots b_m}$ .

1082 We can apply:

- 1083 ■  $A(\alpha_i, 1) \rightarrow A$  for each  $i < j$  until we get  $A(\alpha_j, 0)(\alpha_{j+1}, b_{j+1}) \dots (\alpha_m, b_m) \perp$ ,
- 1084 ■ then apply  $A(\alpha_j, 0) \rightarrow B$  and  $B \rightarrow Z(\alpha_j, 1)$  to get  $Z(\alpha_j, 1)(\alpha_{j+1}, b_{j+1}) \dots (\alpha_m, b_m) \perp$ ,
- 1085 ■ and  $Z \rightarrow Z(\alpha_i, 0)$  for each  $i < j$ , in decreasing order, until we obtain  
 1086  $Z(\alpha_1, 0) \dots (\alpha_{j-1}, 0)(\alpha_j, 1)(\alpha_{j+1}, b_{j+1}) \dots (\alpha_m, b_m) \perp$ , which produces  $\mathbf{a}^{2^{2^m - M - 1}}$  by in-  
 1087 duction hypothesis.

1088 The induction is proven. To obtain the lemma, it suffices to start with  $S$ , apply  $S \rightarrow Z\perp$   
 1089 and then  $Z \rightarrow Z(\alpha_i, 0)$  for each  $i \in [1, k]$  in decreasing order. We get  $Z(\alpha_1, 0) \cdots (\alpha_k, 0)$ ,  
 1090 which produces  $a^{\exp_3(k)}$  by applying the induction with  $M = 0$ . ◀

1091 ▶ **Lemma H.2.** *For all  $E \in N_k$  and  $z \in I_k^*$  the language  $L_\emptyset(Ez)$  contains at most one word.*  
 1092 *In particular,  $L(\mathcal{G}_k)$  is empty or a singleton.*

1093 **Proof.** We prove this for each  $E \in N_k$ , one by one.

- 1094 1. We first observe that from a configuration  $Zz$  there is always at most one rule which  
 1095 can lead to a complete derivation tree: if  $z$  does not contain  $\perp$  then  $L_\emptyset(Zz) = \emptyset$ . If  
 1096  $z = (\alpha_1, b_1) \cdots (\alpha_n, b_n) \perp z'$  then:
  - 1097 ■ if  $n \neq 2^k$  we cannot apply  $Z \xrightarrow{\mathcal{B}_1, \dots, \mathcal{B}_k} D$  since  $z$  has no prefix accepted by all  $\mathcal{B}_i$ .  
 1098 Furthermore, if  $n > 2^k$ , we have  $L_\emptyset(Zz) = \emptyset$  since we can only push more pairs  $(\alpha, 0)$   
 1099 on the stack, so we will never be able to apply  $Z \xrightarrow{\mathcal{B}_1, \dots, \mathcal{B}_k} D$ .  
 1100 ■ if  $n = 2^k$  then we cannot apply  $Z \rightarrow Z(\alpha, 0)$ , by the previous item, as we would obtain  
 1101 more than  $2^k$  symbols before the first  $\perp$ .
- 1102 2. Note that  $B$ ,  $D$  and  $S$  all have a single rule. Meanwhile,  $A$  has several but the top stack  
 1103 symbol determines which rule can be applied. In conclusion, from every configuration  
 1104  $Ez$ , there is at most one rule that can be applied to lead to a complete derivation. As a  
 1105 consequence, the language of  $\mathcal{G}$  contains at most one word.

1106 ◀

1107 By combining the two previous statements we conclude that  $\mathcal{G}_k$  recognises the singleton  
 1108 language  $\{a^{\exp_3(k)}\}$ , while having size only quadratic in  $k$ . A trim NFA for this language  
 1109 must be acyclic, as otherwise it would recognise an infinite language, and thus have at least  
 1110  $\exp_3(k)$  states.